

Traceability Matrix

REMEMBER GORDON! – Rulebook $\rightleftharpoons$ Implementation Mapping

Generated from `docs/traceability.toml`

Overview

Implemented	Descriptive	Implicit	Out-of-scope
87	10	4	9

Total mappings: 110 · Total impl sites: 257

Table of Contents

Overview	2
§1 – Introduction	7
§1.1 – General Comments	7
§1.2 – Game Scale	7
§2 – Game Components	8
§2.1 – The Game Maps	8
§2.2 – Play Aids	8
§2.3 – The Units	8
§2.4 – Game Parts Inventory	9
§2.31 – Dervish weapon types	9
§2.32 – Anglo-Egyptian weapon types	10
§3 – Getting Started	12
§3 – Getting Started	12
§4 – Turn Sequence	13
§4 – Turn Sequence	13
§5 – Movement Phase	15
§5 – Movement Phase (general)	15
§5.3 – Constructing the Zariba	15
§5.11 – Movement allowances printed on units	16
§5.12 – Move up to allowance, hex by hex (cumulative MP cap)	17
§5.13 – No MP accumulation between turns	17
§5.21 – Friendlies transport via gunboat	17
§5.22 – Land units may never enter a Nile River hex	18
§5.23 – Walled city entry restrictions	19
§5.24 – Gunboat upstream/downstream movement	20
§5.25 – Dervish forts may not move	21
§5.26 – Units stop on entering enemy ZOC	21
§5.41 – All units except AE leaders exert ZOC	21
§5.42 – No MP cost to enter/leave enemy ZOC	22
§5.43 – Units stop when entering enemy ZOC	22
§5.44 – ZOC limitations (walls, khor, fort, Nile, Zariba)	23
§5.51 – Stacking limit (4 units + leaders, gunboats isolated)	24
§5.52 – Different Dervish tribes may not stack together	25
§5.53 – Leader stacking with command colour only	25
§5.54 – Anglo-Egyptian Brigade Integrity	26
§6 – Fire Combat Phase	27
§6.3 – Line of Sight Table	27
§6.6 – Special Artillery Capabilities	28
§6.7 – Defensive Fire	28
§6.11 – Fire combat factor printed on units	29
§6.12 – Fire combat is always voluntary	29
§6.13 – Fire factor is unitary (may not be divided)	29
§6.14 – Players may combine fire factors into one attack	30
§6.15 – May divide a stack to fire at different hexes	30
§6.16 – Halving fire strength rounds down, minimum 1	30
§6.21 – First check LOS before firing	30
§6.22 – Consult Range Effects Table	31
§6.23 – Terrain defensive modifier	32
§6.24 – Anglo-Egyptian direct fire accuracy bonus and brigade integrity	32
§6.41 – Direct Fire Subphase	33
§6.42 – Maxim Second Fire and Howitzer Fire Subphase	33
§6.51 – Leader Units	33
§6.52 – Anglo-Egyptian Friendlies Brigade	34
§6.53 – Royal Engineers demolition	34
§6.54 – Forts	35
§6.61 – Only artillery may fire at gunboats; 3+ to sink	36
§6.62 – Only artillery may fire at forts; 2+ to destroy	36
§6.63 – Only artillery may breach wall hexsides; 2+ required	37
§6.64 – Howitzer fire	37
§6.81 – Moving player may fire with all capable units	38

§6.82 – Advance after combat (offensive fire)	38
§7 – Melee Phase	40
§7 – Melee Phase (chapter)	40
§7.1 – Melee strength printed on counter	40
§7.2 – Melee adjacent only, not across wall hexsides	40
§7.3 – Simultaneous melee combat	41
§7.4 – Who may melee attack / defend	41
§7.5 – Cavalry/camel retreat before melee	42
§7.6 – Advance after melee	43
§7.7 – Melee modifiers	43
§8 – Night Game Turns	45
§8.1 – Night effects	45
§8.2 – Dervish Desertion Roll	45
§9 – The Scenarios	47
§9.1 – The Campaign Game	47
§9.2 – The Historical Scenario	47
§9.3 – Bonus Game: Fall of Khartoum	47
§9.11 – Set Up (Campaign)	47
§9.12 – Scenario Length (Campaign)	47
§9.13 – Special Rules (Campaign)	48
§9.14 – Victory Conditions (Campaign)	48
§9.21 – Set Up (Historical)	50
§9.22 – Scenario Length (Historical)	50
§9.23 – Special Rule: The Zariba	50
§9.24 – Victory Conditions (Historical)	51
§9.31 – Bonus game map	52
§9.32 – Set Up (Bonus)	52
§9.33 – Scenario Length (Bonus)	52
§9.34 – Special Rules (Bonus)	52
§9.35 – Victory Conditions (Bonus)	52
§9.111 – Dervish set up (Campaign)	53
§9.112 – Dervish reinforcements (Campaign)	53
§9.113 – Anglo-Egyptian reinforcements (Campaign)	54
§9.211 – Anglo-Egyptian set up first, moves second (Historical)	55
§9.212 – Dervish set up (Historical) – deployment zones and leader-to-hex mapping	55
§9.231 – Thorn hedge hexsides	56
§9.232 – Trench hexsides	56
§9.233 – Zariba entry/exit costs	57
§9.321 – British set up (Bonus)	57
§9.322 – Dervish enters turn one (Bonus)	58
§9.341 – Turn 1 is always a night turn (Bonus)	58
§9.342 – All hexes are playable, including half hexes (Bonus)	59
§9.343 – Both players use the Dervish Range Effects Table (Bonus)	59
§9.344 – Dervish controls the North Fort (Bonus)	59
§9.345 – Gunboat White Nile <-> Blue Nile crossing (Bonus)	60
§9.346 – GORDON immobile, eliminated only at the Palace (Bonus)	60
§10 – Optional Rules	62
§10 – Optional Rules	62
§10.1 – River Mines	62
§10.2 – River Chain	62
§10.11 – Secretly record mine hexes	63
§10.12 – Mine resolution	63
§10.13 – Mines consumed after both rolled for	64
§10.14 – Dervish gunboats pass safely	64
§10.21 – Secretly record chain hexes	64
§10.22 – Gunboat stops on chained hex	65
§10.23 – Sinking the chain	65
§11 – Historical Notes	66
§11 – Historical Notes	66
Credits	67
§Credits – Credits	67
Combat Results Table (shared reference)	68
§CRT – Combat Results Table (shared by §6.22 fire and §7.7 melee)	68

Reference – Charts and Tables	69
§Reference – Charts and Tables	69
Appendix: Symbol Index	70

0/2 implemented

§1 – Introduction

§1.1 – General Comments

descriptive

manual page 1

General Comments

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is a simulation of the final battle in Great Britain’s two-year campaign to reassert her presence in the Sudan (1896–1898). Fought September 2nd, 1898, Omdurman finally broke the back of the fanatical Dervish rebellion and gained Britain a million square miles of desolate territory and two million impoverished subjects. With two players, one assumes the role of Herbert Kitchener, Sirdar (CIC) of the Anglo-Egyptian army; the other player becomes the Khalifa, Abdullah the Taiasha, absolute ruler of the Dervishes. The game is also suited for multi-player participation, with each player assuming command of one or more Dervish tribes or Anglo-Egyptian brigades.

While “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is not, strictly speaking, a beginner’s game, the mechanics of play should be familiar to players of modest experience. It is suggested that the bonus game, FALL OF KHARTOUM, and the historical scenario be played first to familiarize players with the game system prior (see manual for full text)

§1.2 – Game Scale

descriptive

manual page unknown

Game Scale

Each hexagon of the mapsheet represents approximately 400–440 yards of real terrain and each day turn is the equivalent of two hours of real time. Each counter of infantry and cavalry represents between 400 and 700 men, and each of the gunboats present at the battle has its own counter. The upper echelon of command is represented by individual leader counters for the Anglo-Egyptian force; and leaders plus their retinues for the Dervish army.

 3/6 implemented

§2 – Game Components

§2.1 – The Game Maps

descriptive

manual page unknown

The Game Maps

The Omdurman battle map represents approximately 100 square miles of real territory and includes the area north of Omdurman in which the historical battle took place as well as the dominant terrain features that influenced the course of the battle. Note that the mapsheet also contains the Turn Record Track, Turn Sequence, and Terrain Effects Chart at the top; and the Combat Tables and Howitzer Fire Scattergram in the lower right corner. The large letters “A”, “D”, “Y”, etc. are set-up hexes for the historical scenario only (9.2) and should be ignored in the campaign game. Similarly, the hexsides of the Zariba exist only in the historical scenario and should be considered clear terrain in the campaign game. Note, however, that the Anglo-Egyptian player may “construct” the Zariba in the campaign game if desired (see 5.3). All full hexes of the Omdurman game map are playable, including the seven hexes of the Howitzer Fire Scattergram.

The mini-map for the bonus game, FALL OF KHARTOUM, represents that city as it appeared in 1885. The portion (see manual for full text)

See also: §5.3, §9.2

§2.2 – Play Aids

descriptive

manual page unknown

Play Aids

Certain charts and tables are needed to play the game. The Terrain Effects Chart lists all terrain found on the mapsheet and the effect of each type on movement and combat. The Combat Tables describe the range effects on various weapon types and includes the Combat Results Table. Also note the Line of Sight Table on the back of this rulebook. It tells players when certain terrain types block line of sight, thus preventing direct fire attacks on enemy units. Players should become familiar with these various charts and tables prior to the beginning of play.

(see manual for full text)

§2.3 – The Units

implemented

manual page unknown

The Units

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:843 GH:omdurman-types/src/lib.rs:843	UnitKind	<pre>841 /// (`Some(UnitKind::...)`) a non-unit marker counter carries 842 /// `Some(UnitKind::Marker)` or `None`. 843 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 844 pub enum UnitKind { 845 /// Foot infantry (§2.3): fire / melee / movement.</pre>
omdurman-rules/src/	UnitProfile	<pre>779 /// print no melee value). 780 #[derive(Serialize, Deserialize, Clone, Copy,</pre>

lib.rs:781 GH:omdurman-rules/src/lib.rs:781		PartialEq, Eq, Debug)) 781 pub struct UnitProfile { 782 pub kind: UnitKind, 783 pub identity: UnitIdentity,
omdurman-rules/src/lib.rs:16 GH:omdurman-rules/src/lib.rs:16	BrigadeId	14 15 use omdurman_types::{ 16 BrigadeId, BrigadeNationality, DayNight, DervishTribe, Faction, HexCoord, HexsideRef, Player, 17 SetupLetter, UnitKind, 18 };
omdurman-types/src/lib.rs:664 GH:omdurman-types/src/lib.rs:664	SpriteAnnotation	662 /// as an optional overlay over the compiled `sprite_data` fallback. 663 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq)] 664 pub struct SpriteAnnotation { 665 pub color: SpriteColor, 666 pub faction: Option<Faction>,

Covered by tests: [british_army_row_zero_specials_classify_by_counter](#) [egyptian_army_row_zero_specials_classify_by_counter](#)

§2.4 – Game Parts Inventory

out-of-scope

manual page unknown

Game Parts Inventory

Your complete copy of “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN includes:

- One 22 × 28 Battle of Omdurman mapsheet
- One 8½ × 11 bonus game: FALL OF KHARTOUM mapsheet
- One Rules Booklet
- One die-cut Unit Counter Sheet
- One Campaign Game Order of Appearance Card
- One ten-sided die
- One game box

§2.31 – Dervish weapon types

implemented

manual page unknown

Dervish artillery, gunboats, and forts fire on the “artillery” line of the Dervish Range Effects Table; Jehadia and Danagla units fire on the “rifles” line as does the Isa Zachneih unit. All other Dervish units (including leaders) are armed with spears and swords.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:497 GH:omdurman-rules/src/lib.rs:497	WeaponClass	495 /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 496 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 497 pub enum WeaponClass { 498 /// Dervish spears and swords -- no ranged fire at all. 499 Melee,
omdurman-rules/src/unit_profiles.rs:473 GH:omdurman-rules/src/unit_profiles.rs:473	dervish_tribe	471 } 472 473 fn dervish_tribe(tribe: DervishTribe) -> Option<Classification> { 474 // §2.31: "Jehadia and Danagla units fire on the

		'rifles' line as does the 475 // Isa Zachneih unit. All other Dervish units (including leaders) are armed
omdurman- rules/src/unit_profiles.rs:285 GH:omdurman- rules/src/ unit_profiles.rs:285	khalifa_abdullah	283 /// battle (§9.322). All three are interchangeable, so they share the 284 /// `DervishArtillery` identity. 285 fn khalifa_abdullah(col: u32, row: u32) -> Option<Classification> { 286 let artillery = { 287 Some(Classification {
omdurman- rules/src/ lib.rs:998 GH:omdurman- rules/src/lib.rs:998	HowitzerResolution	996 /// roll on the Howitzer Fire Scattergram (§6.64). 997 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 998 pub struct HowitzerResolution { 999 pub combat_results_table_roll: DieRoll, 1000 pub impact_roll: DieRoll,

§2.32 – Anglo-Egyptian weapon types

implemented

manual page unknown

All Anglo-Egyptian units (except gunboats, Maxims, artillery, and leaders) are armed with rifles. Maxims fire on the “Maxims” line of the Anglo-Egyptian Range Effects Table, and artillery and old gunboats fire on the “Artillery” line. New type (named) gunboats may fire on the “Howitzer”, “Artillery”, and “Maxims” lines of the Range Effects Table. (See 6.52 for the fire capabilities of the “Friendlies”).

****Sample Dervish Units**** (printed on counters): combat unit (Combat / Melee / Movement values, plus Tribe identifier); Leader (e.g. OSMAN DIGNA); Camel unit (e.g. Danagla, 4-6-12); Fort.

****Sample Anglo-Egyptian Units**** (printed on counters): Cavalry (e.g. 21 Lancers); Artillery (e.g. 32 Battery); Old Gunboat (e.g. LORD KITCHENER, 0-0-15); New Gunboat — named (e.g. Sultan, with artillery and howitzer factor, plus movement downstream / movement upstream values); Maxim Guns (fire twice per turn); Infantry (Fire Combat Factor / Melee / Movement, plus Battalion ID and Brigade ID — e.g. “2B” = 2nd British Brigade, “3E” = 3rd Egyptian Brigade).

(see manual for full text)

See also: §6.52

File	Symbol	Code Snippet
omdurman- rules/src/ lib.rs:451 GH:omdurman- rules/src/lib.rs:451	GunboatId	449 /// fire; "old" gunboats do not (rulebook §2.32). 450 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 451 pub enum GunboatId { 452 /// One of the five new-type named gunboats with howitzer capability. 453 Named(NamedGunboat),
omdurman- rules/src/ lib.rs:470 GH:omdurman- rules/src/lib.rs:470	NamedGunboat	468 /// The five named gunboats with howitzer capability (rulebook §6.64, §2.32). 469 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 470 pub enum NamedGunboat { 471 Sultan, 472 Melik,
omdurman- rules/src/ lib.rs:483 GH:omdurman- rules/src/lib.rs:483	OldGunboat	481 /// in the Maxim Second Fire and Howitzer subphase (§6.42). 482 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 483 pub enum OldGunboat { 484 LordKitchener, 485 Tamai,

omdurman-rules/src/lib.rs:483 GH:omdurman-rules/src/lib.rs:483	GunboatId::Old	<pre>481 /// in the Maxim Second Fire and Howitzer subphase (\$6.42). 482 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 483 pub enum OldGunboat { 484 LordKitchener, 485 Tamai,</pre>
omdurman-rules/src/lib.rs:457 GH:omdurman-rules/src/lib.rs:457	GunboatId::Dervish-Gunboat	<pre>455 Old(OldGunboat), 456 /// A Dervish gunboat (\$9.111, \$10.14). 457 DervishGunboat(u8), 458 } 459 </pre>

Covered by tests: `old_gunboat_lacks_howitzer` `old_gunboat_rejected_from_howitzer_subphase`

0/1 implemented

| §3 – Getting Started

§3 – Getting Started

out-of-scope

manual page unknown

Getting Started

Spread out the mapsheet on a table. It should lie flat if you backfold it against the scored lines. The Dervish player should sit next to the west edge of the map and the Anglo-Egyptian player opposite him on the east edge. Read through the rules once, looking over the various charts as they are referred to in the various sections. Next, select a scenario and punch out only those unit counters needed to play. Later on, the rest of the unit counters should be punched out, sorted and stored by unit type.

(see manual for full text)

■ 1/1 implemented

§4 – Turn Sequence

§4 – Turn Sequence

implemented

manual page unknown

Turn Sequence

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is played in “Game Turns”, each of which has two “Player Turns”. The player moving first will vary according to the scenario being played. In the Campaign Game, for example, the Anglo-Egyptian player moves first.

A) Anglo-Egyptian Player Turn:

1. Anglo-Egyptian Movement Phase
2. Fire Combat Phase a. Dervish Defensive Fire b. Anglo-Egyptian Offensive Fire
 1. Direct Fire Subphase
 2. Maxim Second Fire and Howitzer Fire Subphase

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:267 GH:omdurman-rules/src/lib.rs:267	GameTurnIndex	<pre>265 /// One-based Game Turn index (1, 2, ... up to the 266 scenario length) (rulebook §4). 267 #[derive(Serialize, Deserialize, Clone, Copy, 268 PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 269 pub struct GameTurnIndex(u8); 270 271 impl GameTurnIndex {</pre>
omdurman-rules/src/lib.rs:297 GH:omdurman-rules/src/lib.rs:297	Phase	<pre>295 /// etc. 296 #[derive(Serialize, Deserialize, Clone, Copy, 297 PartialEq, Eq, Debug, Default)] 298 pub enum Phase { 299 /// Pre-game deployment (§9.2/§9.3/§10): fixed 300 units are placed, each side 301 /// deploys its order of battle within its legal 302 zone, and river</pre>
omdurman-rules/src/effects.rs:806 GH:omdurman-rules/src/effects.rs:806	GameState	<pre>804 /// All mutable state of a game in progress (rulebook 805 §4). 806 #[derive(Serialize, Deserialize, Clone, Debug)] 807 pub struct GameState { 808 pub scenario: Scenario, 809 pub current_turn: GameTurnIndex,</pre>
omdurman-rules/src/effects.rs:942 GH:omdurman-rules/src/effects.rs:942	GameState::new	<pre>940 impl GameState { 941 /// Create a fresh game state for a given scenario 942 (rulebook §4). 943 pub fn new(scenario: Scenario) -> Self { 944 let first = scenario_turn(scenario, 945 GameTurnIndex::new(1)); 946 // First player to *move* per scenario: 947 Campaign -- Anglo-Egyptian moves</pre>
omdurman-rules/src/effects.rs:53 GH:omdurman-rules/src/effects.rs:53	AdvancePhase	<pre>51 /// At end-of-turn, disrupted units recover and 52 per-turn tracking is 53 /// cleared. 54 AdvancePhase, 55 // -- Movement 56 -----</pre>
omdurman-rules/src/effects.rs:2678 GH:omdurman-	advance_phase	<pre>2676 2677 /// Advance the game state to the next phase (rulebook 2678 §4).</pre>

rules/src/ effects.rs:2678		<pre> 2678 pub fn advance_phase(state: &mut GameState) -> Result<(), RuleError> { 2679 let old_phase = state.phase; 2680 debug!(old_phase = ?old_phase, active_player = ? state.active_player, "advance_phase"); </pre>
omdurman- rules/src/ effects.rs:2773 GH:omdurman- rules/src/ effects.rs:2773	end_player_turn	<pre> 2771 2772 /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook §4). 2773 pub fn end_player_turn(state: &mut GameState) -> Result<(), RuleError> { 2774 debug!(2775 old_player = ?state.active_player, </pre>
omdurman- rules/src/ lib.rs:55 GH:omdurman- rules/src/lib.rs:55	GameTurnIn- dex::value	<pre> 53 54 impl \$name { 55 pub fn value(self) -> u16 { 56 match self { 57 \$(Self::\$variant => \$value,)+ </pre>
omdurman- rules/src/ effects.rs:934 GH:omdurman- rules/src/ effects.rs:934	PendingMelee	<pre> 932 /// resolution after the reaction window is deterministic and host-ordered (rulebook §7.5). 933 #[derive(Serialize, Deserialize, Clone, Debug)] 934 pub struct PendingMelee { 935 pub attack: MeleeAttack, 936 pub attacker_roll: DieRoll, </pre>

Covered by tests: both_ready_auto_advances_out_of_setup fire_combat_wrong_phase_rejected new_game_starts_in_setup scenario_turn_dispatches_correctly turn_advances_through_phases game_turn_marker_cell_returns_none

18/19 implemented

§5 – Movement Phase

§5 – Movement Phase (general)

descriptive

manual page unknown

Movement Phase

Covered by tests: `disrupted_unit_cannot_fire` `disrupted_unit_may_not_act`

§5.3 – Constructing the Zariba

implemented

manual page unknown

Constructing the Zariba

The Zariba trench and thorn hedge hexsides are built and in place in the historical scenario only. These hexsides are considered clear terrain in the campaign game. The Anglo-Egyptian player may, however, find it useful to construct this defensive position during the campaign game. The Zariba hexsides may only be built in their position as displayed on the mapsheet. Construction procedure is as follows: any Anglo-Egyptian infantry unit that begins and ends the Anglo-Egyptian player turn adjacent to (and on the Nile side of) Zariba hexsides has constructed all Zariba hexsides to which he is adjacent. The constructing unit may neither fire offensively nor melee attack during the turn of construction. Use a blank counter to denote units constructing Zariba hexsides. See 9.23 for defensive benefits and movement restrictions of Zariba hexsides.

(see manual for full text)

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1735 GH:omdurman-rules/src/lib.rs:1735	constructing_zariba	<pre>1733 // melee attack during the turn of 1734 construction." 1735 let s = UnitState { 1736 constructing_zariba: true, 1737 ..UnitState::default() 1738 };</pre>
omdurman-rules/src/effects.rs:205 GH:omdurman-rules/src/effects.rs:205	ConstructZariba	<pre>203 /// Begin constructing a Zariba hexside (rulebook 204 §5.3). 205 ConstructZariba { 206 unit_ids: Vec<UnitId>, 207 hexside: HexsideRef,</pre>
omdurman-rules/src/effects.rs:4418 GH:omdurman-rules/src/effects.rs:4418	apply_construct_zariba	<pre>4416 /// Mark a set of units as constructing a Zariba 4417 hexside (rulebook §5.3). 4418 pub fn apply_construct_zariba(4419 state: &mut GameState, 4420 unit_ids: &[UnitId],</pre>
omdurman-rules/src/lib.rs:834 GH:omdurman-rules/src/lib.rs:834	UnitState::may_attack_t	<pre>832 /// A unit that began construction this turn may 833 not fire offensively or 834 /// melee (§5.3, §6.53). 835 pub fn may_attack_this_turn(self) -> bool { 836 !self.disrupted && !self.constructing_zariba 837 && !self.demolishing 838 }</pre>

§5.11 – Movement allowances printed on units

implemented

manual page unknown

The movement allowances of the various unit types are printed directly on the units (see 2.3). A unit may move up to this printed movement allowance, paying varying costs for different terrain types (see the Terrain Effects Chart).

See also: §2.3

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:123 GH:omdurman-rules/src/lib.rs:123	MovementAllowance	<pre>121 /// is a named variant. 122 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 123 pub enum MovementAllowance { 124 /// Immobile (forts, wrecked gunboats). 125 Immobile = 0,</pre>
omdurman-rules/src/lib.rs:792 GH:omdurman-rules/src/lib.rs:792	UnitMovement	<pre>790 /// Movement allowance -- uniform for land units, split for gunboats (rulebook §5.11, §5.24, §5.25). 791 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 792 pub enum UnitMovement { 793 Land(MovementAllowance), 794 Gunboat(GunboatMovement),</pre>
omdurman-types/src/lib.rs:291 GH:omdurman-types/src/lib.rs:291	HexDirection	<pre>289 /// (`+q`, `+q+r`, `+r`, `-q`, `-q-r`, `-r` for pointy- top hexes) (rulebook §5.11, §5.24). 290 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, Default)] 291 pub enum HexDirection { 292 #[default] 293 East = 0,</pre>
omdurman-rules/src/lib.rs:165 GH:omdurman-rules/src/lib.rs:165	MovementPoints	<pre>163 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, Default, 164)] 165 pub struct MovementPoints(i16); 166 167 impl MovementPoints {</pre>
omdurman-rules/src/terrain_chart.rs:21 GH:omdurman-rules/src/terrain_chart.rs:21	terrain_effects_chart	<pre>19 /// 20 /// Source: printed Terrain Effects Chart on the mapsheet. 21 pub fn terrain_effects_chart(terrain: Terrain) -> TerrainEntry { 22 match terrain { 23 Terrain::Clear { .. } => TerrainEntry {</pre>
omdurman-rules/src/terrain_chart.rs:77 GH:omdurman-rules/src/terrain_chart.rs:77	movement_cost	<pre>75 /// Convenience: get the movement cost for a terrain type (rulebook §5.11, Terrain Effects Chart). 76 /// Returns `None` for impassable terrain (Nile). 77 pub fn movement_cost(terrain: Terrain) -> Option<MovementAllowance> { 78 terrain_effects_chart(terrain).movement_cost 79 }</pre>
omdurman-rules/src/terrain_chart.rs:85 GH:omdurman-rules/src/terrain_chart.rs:85	movement_cost_with_road	<pre>83 /// underlying terrain; without a road it's the terrain's own cost. The road is 84 /// a movement overlay only -- combat/LOS still use the underlying terrain. 85 pub fn movement_cost_with_road(terrain: Terrain, road: bool) -> Option<MovementAllowance> { 86 if road { 87 Some(MovementAllowance::One)</pre>
omdurman-types/src/lib.rs:422 GH:omdurman-types/src/lib.rs:422	Terrain::passable_by_land	<pre>420 421 /// Whether this terrain may be entered by land units (rulebook §5.11). 422 pub fn passable_by_land(self) -> bool {</pre>

		<pre> 423 !self.is_nile() 424 } </pre>
omdurman-types/src/lib.rs:488 GH:omdurman-types/src/lib.rs:488	Terrain::is_crossroad	<pre> 486 487 /// Whether roads converge at this hex's centre. 488 pub fn is_crossroad(self) -> bool { 489 matches!(self.road(), Road::Crossroad) 490 } </pre>
omdurman-hexmap/src/map.rs:17 GH:omdurman-hexmap/src/map.rs:17	GameMap::roads	<pre> 15 pub hexes: HashMap<HexCoord, HexData>, 16 pub hexsides: HashMap<HexsideRef, HexsideKind>, 17 pub roads: HashSet<HexsideRef>, 18 pub excluded: HashSet<HexCoord>, 19 pub overlay: OverlayParams, </pre>

Covered by tests: [clear_terrain_no_bonus](#) [nile_is_impassable](#) [rough_movement_and_defense](#) [swamp_movement_and_defense](#) [hilltop_movement_and_defense](#) [huts_movement_and_defense](#) [movement_cost_convenience_matches_chart](#) [movement_cost_with_road_always_one](#) [land_unit_may_not_enter_nile](#) [movement_cost_without_road_matches_terrain](#) [movement_cost_for_uses_terrain](#) [movement_cost_for_road_costs_one](#) [road_gives_crossroad](#) [terrain_movement_costs_in_bounds](#) [terrain_chart_road_always_costs_one](#)

§5.12 – Move up to allowance, hex by hex (cumulative MP cap)

implemented

manual page unknown

A player may move as many or as few of his units as desired during each movement phase, limited only by the units' movement allowance, the terrain costs paid in moving from hex to hex, and enemy zones of control (see 5.4).

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:2158 GH:omdurman-rules/src/effects.rs:2158	mp_spent	<pre> 2156 2157 /// Movement points `unit_id` has already spent 2158 this turn (§5.11/§5.12). 2159 pub fn mp_spent(&self, unit_id: UnitId) -> i16 { 2160 self.mp_spent_this_turn.get(&unit_id).copied().unwrap_or(0) </pre>

§5.13 – No MP accumulation between turns

implemented

manual page unknown

A unit may never accumulate movement points from turn to turn, nor may a unit transfer unused movement points to other units. A unit's unused movement points in any given turn are considered lost.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:2773 GH:omdurman-rules/src/effects.rs:2773	end_player_turn	<pre> 2771 2772 /// End the current player's turn: recover disrupted 2773 units, switch active player, advance turn index (rulebook 2774 §4). 2775 pub fn end_player_turn(state: &mut GameState) -> 2776 Result<(), RuleError> { 2777 debug!(2778 old_player = ?state.active_player, </pre>

§5.21 – Friendlies transport via gunboat

implemented

manual page unknown

In general, naval transport missions are not allowed, i.e. gunboats may not carry any land units. The sole exception is that the Anglo-Egyptian player may transport the surviving units of the “Friendlies” brigade from the east bank of the Nile to the west bank after, and only after, the Dervish east bank unit (Isa Zachneih) has been eliminated. The transport is accomplished in the following sequence: a) on any turn that a “Friendlies” unit and any Anglo-Egyptian gunboat start their turn adjacent, that unit may load onto (i.e. stack with) the gunboat; b) during the Anglo-Egyptian player’s next turn the gunboat may move to any Nile hex adjacent to a west bank hex (up to the gunboat’s movement allowance); c) on the Anglo-Egyptian player’s third turn the “Friendlies” unit may disembark and move normally, paying the normal terrain cost for the first hex entered. The gunboat may also move normally that turn.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:631 GH:omdurman-rules/src/lib.rs:631	is_friendlylies	<pre> 629 /// "Friendlies" units obey several special rules (\$5.21, \$5.23, \$6.52, 630 /// \$9.14 victory conditions). 631 pub fn is_friendlylies(&self) -> bool { 632 matches!(633 self, </pre>
omdurman-rules/src/lib.rs:812 GH:omdurman-rules/src/lib.rs:812	loaded_on	<pre> 810 pub disrupted: bool, 811 /// `Some(gunboat)` after a "Friendlies" unit loads onto a gunboat (\$5.21). 812 pub loaded_on: Option<UnitId>, 813 /// Set while the unit is building Zariba hexsides -- neither offensive 814 /// fire nor melee allowed that turn (\$5.3). </pre>
omdurman-rules/src/effects.rs:232 GH:omdurman-rules/src/effects.rs:232	FriendliesTransport	<pre> 230 231 /// Load/disembark the "Friendlies" brigade via gunboat (rulebook \$5.21). 232 FriendliesTransport(crate::FriendliesAction), 233 234 // -- Optional rules ----- </pre>
omdurman-rules/src/effects.rs:4848 GH:omdurman-rules/src/effects.rs:4848	apply_friendlylies_transpo	<pre> 4846 /// unit is freed (a disembarking `MoveUnit` should follow, costed by 4847 /// terrain). 4848 pub fn apply_friendlylies_transport(4849 state: &mut GameState, 4850 action: FriendliesAction, </pre>
omdurman-rules/src/lib.rs:1074 GH:omdurman-rules/src/lib.rs:1074	FriendliesAction	<pre> 1072 /// tracks each unit–gunboat pair independently. 1073 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1074 pub enum FriendliesAction { 1075 /// Turn N (the load turn): unit and gunboat started adjacent; unit 1076 /// loads onto (stacks with) the gunboat. </pre>
omdurman-rules/src/lib.rs:1090 GH:omdurman-rules/src/lib.rs:1090	TransportState	<pre> 1088 /// third turn. 1089 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1090 pub enum TransportState { 1091 /// Turn N (the load turn): unit and gunboat started adjacent; unit 1092 /// loads onto (stacks with) the gunboat. </pre>

§5.22 – Land units may never enter a Nile River hex

implemented

manual page unknown

With the exception of 5.21, land units may never enter a Nile River hex. Only gunboats may enter and move along Nile River hexes.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1476 GH:omdurman-rules/src/effects.rs:1476	can_move_unit_to	<pre> 1474 /// 1475 /// [<code>`hex_in_enemy_zoc`</code>]: Self::hex_in_enemy_zoc 1476 pub fn can_move_unit_to(1477 &self, 1478 unit_id: UnitId,</pre>
omdurman-rules/src/effects.rs:1154 GH:omdurman-rules/src/effects.rs:1154	in_deployment_zone	<pre> 1152 /// plan / UI rather than this hex predicate. 1153 Documented, not silently 1154 /// dropped. 1155 pub fn in_deployment_zone(&self, player: Player, 1156 hex: HexCoord, is_boat: bool) -> bool { 1157 // No board attached -> permissive (unit 1158 tests, unbound session). 1159 if self.board.terrain.is_empty() {</pre>

Covered by tests: [campaign_deployment_is_boat_land_exclusive](#) [fok_ae_gunboat_deploys_only_on_nile](#) [fok_ae_land_unit_rejected_on_nile](#) [deploy_via_real_sprite_resolution_matches_engine](#) [fok_dervish_land_unit_rejected_on_nile](#) [retreat_before_melee_may_not_land_on_nile](#)

§5.23 – Walled city entry restrictions

implemented

manual page unknown

Only certain units may enter the walled portion of Omdurman. For the Dervish player these are the Khalifa unit, the three Dervish artillery units, and the Taiasha units (the Khalifa's bodyguard). Any Anglo-Egyptian units that can get to the walled city may enter it (except gunboats and "Friendlies"). Units entering and/or exiting the walled city may only do so through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:141 GH:omdurman-types/src/lib.rs:141	HexsideRef	<pre> 139 /// data by [<code>`HexsideRef`</code>]. 140 #[derive(Serialize, Deserialize, Clone, Copy, 141 PartialEq, Eq, Hash, Debug)] 142 pub struct HexsideRef { 143 pub a: HexCoord,</pre>
omdurman-types/src/lib.rs:176 GH:omdurman-types/src/lib.rs:176	HexsideKind	<pre> 174 strum::EnumIter, 175)] 176 pub enum HexsideKind { 177 /// City wall (Khartoum, walled city of Omdurman). 178 Blocks LOS, blocks 179 /// movement except across gates/breaches (§5.23), 180 blocks ZOC into the city</pre>
omdurman-types/src/lib.rs:247 GH:omdurman-types/src/lib.rs:247	blocks_movement	<pre> 245 /// `omdurman-rules`. The trench *end* variants 246 are therefore intentionally 247 /// not blocking. 248 pub fn blocks_movement(self) -> bool { 249 matches!(250 self,</pre>
omdurman-rules/src/lib.rs:657 GH:omdurman-rules/src/lib.rs:657	may_enter_walled_city	<pre> 655 /// Taiasha bodyguard may enter. Anglo-Egyptian: 656 any unit that can reach the 657 /// walled city *except* gunboats and "Friendlies". 658 pub fn may_enter_walled_city(&self) -> bool { 659 match self {</pre>

		<pre> 659 // \$5.23 Dervish: Khalifa, artillery, Taiasha. </pre>
omdurman-rules/src/board.rs:294 GH:omdurman-rules/src/board.rs:294	is_walled_city	<pre> 292 /// touches it on one. The two-sided threshold keeps the predicate robust to 293 /// a map edit that adds or removes a single wall segment. 294 pub fn is_walled_city(&self, hex: HexCoord) -> bool { 295 // Membership in the precomputed enclosed area (see `walled_city`). 296 // Palace/Tomb hexes are always part of it (they are the seeds). </pre>
omdurman-rules/src/effects.rs:405 GH:omdurman-rules/src/effects.rs:405	WalledCityEntry	<pre> 403 404 #[error("unit {0:?} is not eligible to enter the walled city of Omdurman at {1:?} (§5.23)")] 405 WalledCityEntry(UnitId, HexCoord), 406 407 #[error("movement cost {cost:?} exceeds allowance {allowance:?}")] </pre>

Covered by tests: [can_move_rejects_wall_hexside](#) [can_move_allows_gate_hexside](#) [walled_city_entry_allows_khalifa](#) [walled_city_entry_rejects_unauthorized_dervish](#) [walled_city_entry_rejects_ae_gunboat](#) [walled_city_entry_not_enforced_for_fok](#) [campaign_walled_city_is_enclosed_by_walls](#)

§5.24 – Gunboat upstream/downstream movement

implemented

manual page unknown

Note that gunboats have two movement allowances separated by a slash, e.g. 10/16. The smaller number is the movement allowance when moving upstream, i.e. against the current (the direction of the current is indicated by arrows in the Nile). The larger number is the movement allowance when moving downstream, i.e. with the current. Gunboats may combine movement in both directions, but if they move even one hex upstream, their upstream movement allowance is their maximum movement allowance for that turn, and may not be exceeded.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:565 GH:omdurman-rules/src/lib.rs:565	GunboatMovement	<pre> 563 /// the turn. 564 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 565 pub struct GunboatMovement { 566 pub upstream: MovementAllowance, 567 pub downstream: MovementAllowance, </pre>
omdurman-types/src/lib.rs:900 GH:omdurman-types/src/lib.rs:900	is_boat	<pre> 898 } 899 900 /// Gunboats use the split upstream/downstream movement allowance (§5.24). 901 pub fn is_boat(self) -> bool { 902 matches!(self, UnitKind::Gunboat { .. }) </pre>

Covered by tests: [boat_annotation_yields_split_gunboat_movement](#) [gunboat_upstream_cap_is_sticky_across_moves](#)

§5.25 – Dervish forts may not move

implemented

manual page unknown

Dervish forts may not move in any way once placed.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:796 GH:omdurman-rules/src/lib.rs:796	Immobile	<pre>794 Gunboat(GunboatMovement), 795 /// Forts may not move once placed (§5.25). 796 Immobile, 797 } 798 </pre>
omdurman-types/src/lib.rs:861 GH:omdurman-types/src/lib.rs:861	UnitKind::Fort	<pre>859 Gunboat { fire: i32, upstream: i32, downstream: 860 i32 }, 861 /// Permanent emplacement (§6.54): fire 862 (artillery) / melee (defensive). 863 /// May not move once placed (§5.25). 864 Fort { fire: i32, melee: i32 }, 865 /// Dervish leader (§6.51): fire / melee / 866 movement. May melee attack (§7.4).</pre>
omdurman-rules/src/lib.rs:796 GH:omdurman-rules/src/lib.rs:796	UnitMovement::Immobile	<pre>794 Gunboat(GunboatMovement), 795 /// Forts may not move once placed (§5.25). 796 Immobile, 797 } 798 </pre>

Covered by tests: [forts_are_never_advance_eligible](#)

§5.26 – Units stop on entering enemy ZOC

implemented

manual page unknown

Units must stop their movement immediately upon entering an enemy zone of control (see 5.4).

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1476 GH:omdurman-rules/src/effects.rs:1476	can_move_unit_to	<pre>1474 /// 1475 /// [<code>hex_in_enemy_zoc`</code>]: Self::<code>hex_in_enemy_zoc</code> 1476 pub fn can_move_unit_to(1477 &self, 1478 unit_id: UnitId,</pre>
omdurman-rules/src/effects.rs:2329 GH:omdurman-rules/src/effects.rs:2329	hex_in_enemy_zoc	<pre>2327 /// does not extend into or out of a Nile hex. 2328 With no board loaded these 2329 /// reduce to the plain adjacency rule. 2330 pub fn hex_in_enemy_zoc(2331 &self, 2332 hex: HexCoord,</pre>

Covered by tests: [unit_entering_enemy_zoc_may_move_no_further_that_turn](#) [zoc_transit_check_uses_the_actual_path](#)

§5.41 – All units except AE leaders exert ZOC

implemented

manual page unknown

All units except Anglo-Egyptian leaders exert a zone of control (hereafter called a ZOC) into their six adjacent hexes (exception: Gunboats exert a ZOC only against enemy gunboats). Disrupted units have no ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:864 GH:omdurman-rules/src/lib.rs:864	ZocReason	<pre> 862 /// Used by the engine when answering "is this hex in an enemy ZOC?". 863 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 864 pub enum ZocReason { 865 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 866 /// leader (§5.41) projects ZOC into each of its six adjacent hexes. </pre>
omdurman-rules/src/effects.rs:2294 GH:omdurman-rules/src/effects.rs:2294	unit_projects_zoc	<pre> 2292 /// §5.44) need the game map, which the engine does not hold; the app layers 2293 /// those on top. This is the position/kind/ disruption core of the rule. 2294 pub fn unit_projects_zoc(2295 &self, 2296 unit: &UnitPlacement, </pre>
omdurman-rules/src/effects.rs:2370 GH:omdurman-rules/src/effects.rs:2370	zoc_hexes	<pre> 2368 /// ZOC covers a given hex; this function returns *which* hexes a 2369 /// specific unit covers. 2370 pub fn zoc_hexes(&self, unit: &UnitPlacement, mover_player: Player, mover_kind: UnitKind) -> Vec<HexCoord> { 2371 let Some(reason) = self.unit_projects_zoc(unit, mover_player, mover_kind) else { 2372 return Vec::new(); </pre>

Covered by tests: `zoc_hexes_empty_for_anglo_egyptian_leader` `zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions` `zoc_hexes_empty_for_disrupted_unit` `zoc_hexes_matches_hex_in_enemy_zoc` `zoc_hexes_excludes_nile` `zoc_hexes_excludes_khor`

§5.42 – No MP cost to enter/leave enemy ZOC

implemented

manual page unknown

There is no movement point cost to enter or leave an enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1639 GH:omdurman-rules/src/effects.rs:1639	movement_cost_for	<pre> 1637 /// 1638 /// §5.42: entering or leaving an enemy ZOC adds no MP cost. 1639 fn movement_cost_for(&self, unit: &UnitPlacement, path: &[HexCoord]) -> Option<MovementPoints> { 1640 if path.is_empty() self.board.terrain.is_empty() { 1641 return None; </pre>

Covered by tests: `entering_enemy_zoc_costs_no_extra_mp`

§5.43 – Units stop when entering enemy ZOC

implemented

manual page unknown

All units must stop when they enter an enemy ZOC and may move no further that turn. In their next movement phase they may withdraw or, if desired, move directly into another enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/	can_move_unit_to	<pre> 1474 /// 1475 /// [<code>hex_in_enemy_zoc</code>]: Self::hex_in_enemy_zoc </pre>

effects.rs:1476 GH:omdurman-rules/src/effects.rs:1476		1476 pub fn can_move_unit_to(1477 &self, 1478 unit_id: UnitId,
omdurman-rules/src/effects.rs:2329 GH:omdurman-rules/src/effects.rs:2329	hex_in_enemy_zoc	2327 /// does not extend into or out of a Nile hex. With no board loaded these 2328 /// reduce to the plain adjacency rule. 2329 pub fn hex_in_enemy_zoc(2330 &self, 2331 hex: HexCoord,

Covered by tests: `unit_entering_enemy_zoc_may_move_no_further_that_turn`

§5.44 – ZOC limitations (walls, khor, fort, Nile, Zariba)

implemented

manual page unknown

ZOCs do not extend into or out of a Nile River hex (exception: Gunboats, see 5.41). ZOCs do not extend across a khor, into a fort, or into a hex inside the walled city across a wall hexside. ZOCs do extend out of a fort (even if unoccupied), and from a walled city hex into an adjacent non-walled-city hex across a wall hexside. ZOCs also extend out of (but not into) a walled city hex across a gate hexside. ZOCs extend both ways across a breach hexside. ZOCs also extend out of, but not into, a hut or building hex. In the historical scenario ZOCs extend out of, but not into, the Zariba across a Zariba hexside (also in the campaign game if the Zariba is constructed).

(see manual for full text)

See also: §5.41

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:864 GH:omdurman-rules/src/lib.rs:864	ZocReason	862 /// Used by the engine when answering "is this hex in an enemy ZOC?". 863 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 864 pub enum ZocReason { 865 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 866 /// leader (§5.41) projects ZOC into each of its six adjacent hexes.
omdurman-types/src/lib.rs:181 GH:omdurman-types/src/lib.rs:181	Wall	179 /// (§5.44), blocks melee (§7.2), blocks advance- after-combat (§6.82). 180 #[default] 181 Wall, 182 /// Gate hexside in a wall. ZOC extends *out of* the walled city through 183 /// gates but not into it (§5.44). Melee may be made through a gate (§7.2).
omdurman-types/src/lib.rs:190 GH:omdurman-types/src/lib.rs:190	Khor	188 /// Khor -- gully/wadi. ZOCs do not extend across (§5.44); advance after 189 /// combat may not cross (§6.82). 190 Khor, 191 /// Crest line. Blocks LOS unless the firer is on the higher side 192 /// (§6.3 note 7).
omdurman-rules/src/lib.rs:877	ZocReason::Zariba	875 /// Zariba hexside ZOC behaviour in the historical scenario / when the 876 /// Zariba is constructed (§5.44). 877 Zariba,

GH:omdurman-rules/src/lib.rs:877		878 } 879
omdurman-rules/src/effects.rs:2294 GH:omdurman-rules/src/effects.rs:2294	unit_projects_zoc	2292 /// §5.44) need the game map, which the engine does not hold; the app layers 2293 /// those on top. This is the position/kind/disruption core of the rule. 2294 pub fn unit_projects_zoc(2295 &self, 2296 unit: &UnitPlacement,
omdurman-types/src/lib.rs:264 GH:omdurman-types/src/lib.rs:264	HexsideKind::blocks_zoc	262 /// cannot express; those are left to the caller. This predicate captures the 263 /// symmetric "does not extend across" cases. 264 pub fn blocks_zoc(self) -> bool { 265 matches!(266 self,
omdurman-rules/src/effects.rs:2329 GH:omdurman-rules/src/effects.rs:2329	hex_in_enemy_zoc	2327 /// does not extend into or out of a Nile hex. With no board loaded these 2328 /// reduce to the plain adjacency rule. 2329 pub fn hex_in_enemy_zoc(2330 &self, 2331 hex: HexCoord,

Covered by tests: `zoc_hexes_excludes_nile` `zoc_hexes_excludes_khor` `zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions`

§5.51 – Stacking limit (4 units + leaders, gunboats isolated)

implemented

manual page unknown

No more than four units may occupy a hex, with the exception of leaders and gunboats. All leader units are free stacking, i.e. they may stack in addition to the four-unit-per-hex stacking limitation. Gunboats may not stack with any other unit (Exception: 5.21). Players may move through friendly units at no additional cost in movement points. The stacking limitation applies only at the end of the movement phase and during combat.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:886 GH:omdurman-rules/src/lib.rs:886	OverLimit	884 /// and the gunboat exception. 885 #[error("hex stack exceeds the four-unit limit")] 886 OverLimit, 887 /// "Gunboats may not stack with any other unit" (§5.51, exception §5.21). 888 #[error("gunboats may not stack with non-gunboat units")]
omdurman-rules/src/lib.rs:889 GH:omdurman-rules/src/lib.rs:889	GunboatStack	887 /// "Gunboats may not stack with any other unit" (§5.51, exception §5.21). 888 #[error("gunboats may not stack with non-gunboat units")] 889 GunboatStack, 890 /// "Units of different Dervish tribes may not stack together" (§5.52). 891 #[error("Dervish units of different tribes may not stack")]
omdurman-rules/src/effects.rs:2214 GH:omdurman-rules/src/effects.rs:2214	check_stacking	2212 /// * §5.52 -- units of different Dervish tribes may not stack together. 2213 /// * §5.53 -- a Dervish leader may stack only with units of its command. 2214 pub fn check_stacking(2215 &self, 2216 mover: &UnitPlacement,

omdurman-rules/src/effects.rs:2395 GH:omdurman-rules/src/effects.rs:2395	check_invariants	<pre>2393 /// violations (empty = valid). Useful as a 2394 postcondition assertion 2395 /// after `apply_effect`. 2396 pub fn check_invariants(&self) -> Vec<&'static 2397 str> { 2398 let mut violations = Vec::new();</pre>
---	-------------------------	--

Covered by tests: [stacking_over_limit_rejected](#) [mid_move_stacking_allows_pass_through](#)
[mid_move_stacking_rejects_over_limit_destination](#) [check_invariants_clean_state](#) [check_invariants_catches_stacking_violation](#)
[check_invariants_allows_leaders_stacking](#)

§5.52 – Different Dervish tribes may not stack together

[implemented](#)
manual page unknown

The units of different Dervish tribes may not stack together, even if they are the same color (e.g. although both are green, Mulazmin and Jehadia units may not stack with each other).

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:892 GH:omdurman-rules/src/lib.rs:892	DervishTribeMix	<pre>890 /// "Units of different Dervish tribes may not 891 stack together" (§5.52). 892 #[error("Dervish units of different tribes may not 893 stack")] 894 DervishTribeMix, 895 /// "If Dervish leaders elect to stack, they may 896 only stack with units of 897 their command (i.e. colour)" (§5.53).</pre>
omdurman-rules/src/effects.rs:2395 GH:omdurman-rules/src/effects.rs:2395	check_invariants	<pre>2393 /// violations (empty = valid). Useful as a 2394 postcondition assertion 2395 /// after `apply_effect`. 2396 pub fn check_invariants(&self) -> Vec<&'static 2397 str> { 2398 let mut violations = Vec::new();</pre>

Covered by tests: [green_sections_are_mulazmin_tribal_units](#) [deploy_rejects_dervish_tribe_mix](#) [check_invariants_clean_state](#)

§5.53 – Leader stacking with command colour only

[implemented](#)
manual page unknown

Leader units are not required to stack. If Dervish leaders elect to stack, however, they may only stack with units of their command (i.e. color). For example, Sheik El Din may only stack with Mulazmins or Jehadias.

****5.54) Anglo-Egyptian Brigade Integrity:**** All British, Sudanese, and Egyptian infantry units have their brigade designation printed in the upper right corner (e.g. “2B” = 2nd British Brigade; “3E” = 3rd Egyptian Brigade, etc.). In any combat phase in which all four infantry battalions belonging to any Anglo-Egyptian infantry brigade are stacked in the same hex they are said to have brigade integrity. Stacks having brigade integrity receive a +1 modifier to their fire combat die roll provided they all fire at the same enemy occupied hex. This modifier is in addition to the normal +1 bonus given to all Anglo-Egyptian direct fire attacks (see 6.24).

(see manual for full text)

See also: §5.54, §6.24

File	Symbol	Code Snippet
------	--------	--------------

omdurman-rules/src/lib.rs:896 GH:omdurman-rules/src/lib.rs:896	DervishLeaderCommandMismatch	<pre> 894 /// their command (i.e. colour)" (\$5.53). 895 #[error("Dervish leader may only stack with units of their own command")] 896 DervishLeaderCommandMismatch, 897 } 898 </pre>
---	-------------------------------------	---

§5.54 – Anglo-Egyptian Brigade Integrity

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:902 GH:omdurman-rules/src/lib.rs:902	BrigadeIntegrity	<pre> 900 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 901 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 902 pub enum BrigadeIntegrity { 903 None, 904 Integrated(BrigadeId), </pre>
omdurman-rules/src/lib.rs:754 GH:omdurman-rules/src/lib.rs:754	brigade_integrity	<pre> 752 /// Only a full stack of four battalions qualifies. Three or fewer may still 753 /// stack and fire, but they receive no brigade- integrity bonus. 754 pub fn brigade_integrity(identities: &[UnitIdentity]) - > BrigadeIntegrity { 755 let Some(brigade) = identities.first().and_then(i i.brigade()) else { 756 return BrigadeIntegrity::None; </pre>
omdurman-rules/src/lib.rs:902 GH:omdurman-rules/src/lib.rs:902	FireModifier::BrigadeIntegrity	<pre> 900 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 901 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 902 pub enum BrigadeIntegrity { 903 None, 904 Integrated(BrigadeId), </pre>
omdurman-rules/src/lib.rs:352 GH:omdurman-rules/src/lib.rs:352	BattalionOrdinal	<pre> 350 /// brigade integrity requires all four stacked in one hex (\$5.54). 351 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 352 pub enum BattalionOrdinal { 353 First = 1, 354 Second = 2, </pre>
omdurman-types/src/lib.rs:959 GH:omdurman-types/src/lib.rs:959	BrigadeId	<pre> 957 /// (\$5.54 enumerates only British/Egyptian/Sudanese) but ride along on the 958 /// same field for uniform handling. 959 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 960 pub struct BrigadeId { 961 pub number: u8, </pre>

Covered by tests: [brigade_designation_ignored_for_non_infantry](#) [printed_brigade_designation_overrides_column](#)
[tribe_stats_come_from_annotation](#) [section_owner_anglo_egyptian_sections](#) [section_owner_green_sections_are_dervish](#)
[brigade_integrity_four_battalions_returns_integrated](#) [ae_infantry_fourth_battalion_from_col_3](#) [brigade_integrity_empty_slice](#)
[brigade_integrity_friendlys_returns_none](#) [section_owner_dervish_sections](#) [brigade_integrity_three_battalions_returns_none](#)
[unit_identity_brigade_and_battalion_accessors](#) [ae_infantry_brigade_number_three_from_col_7](#)
[brigade_integrity_non_infantry_returns_none](#) [brigade_integrity_mixed_brigades_returns_none](#)
[ae_infantry_third_battalion_from_col_2](#) [brigade_integrity_modifier_is_engine_derived](#)

21/25 implemented

§6 – Fire Combat Phase

§6.3 – Line of Sight Table

implemented

manual page unknown

Line of Sight Table

This table is located on the back of this rulebook and should be self-explanatory. Locate the terrain type the firing unit is in and cross-index it with the terrain type the target unit is in. Terrain types in the intersecting box block line of sight, with exceptions as footnoted. Also study the “Special LOS Notes” given and remember that (with the exception of howitzer fire — see 6.64) you can’t fire at anything you can’t see!

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:55 GH:omdurman-rules/src/los_table.rs:55	LosLevel	<pre>53 /// Ordered lowest to highest: `Ground < Rough < 54   Hilltop`. 55 #[derive(Clone, Copy, PartialEq, Eq, PartialOrd, Ord, 56 Hash, Debug)] 57 pub enum LosLevel { Ground, Rough,</pre>
omdurman-rules/src/los_table.rs:63 GH:omdurman-rules/src/los_table.rs:63	LosFeature	<pre>61 /// A feature on the LOS ray that may block (rulebook 62 §6.3). 63 #[derive(Clone, Copy, PartialEq, Eq, Hash, Debug)] 64 pub enum LosFeature { 65 /// A hex containing units (gunboats/forts excluded per note a). Units,</pre>
omdurman-rules/src/los_table.rs:83 GH:omdurman-rules/src/los_table.rs:83	LosCondition	<pre>81 /// A positional condition from the LOS table Detail 82 footnotes. 83 #[derive(Clone, Copy, PartialEq, Eq, Debug)] 84 pub enum LosCondition { 85 /// (1) Blocks only if the ray passes through more than two such features. MoreThanTwo,</pre>
omdurman-rules/src/los_table.rs:121 GH:omdurman-rules/src/los_table.rs:121	los_level	<pre>119 120 /// Map a terrain type to its LOS level (rulebook 121 §6.3). 122 pub fn los_level(terrain: Terrain) -> LosLevel { 123 use LosLevel::*; 124 match terrain {</pre>
omdurman-rules/src/los_table.rs:139 GH:omdurman-rules/src/los_table.rs:139	los_level_for_unit	<pre>137 /// 138 /// For all other units, the level is derived from the 139 terrain at `hex`. 140 pub fn los_level_for_unit(141 kind: UnitKind, 142 hex: HexCoord,</pre>
omdurman-rules/src/los_table.rs:175 GH:omdurman-rules/src/los_table.rs:175	blocking_rules	<pre>173 /// if ALL conditions are satisfied (AND semantics). An 174 empty conditions slice 175 /// means the feature always blocks. 176 pub fn blocking_rules(177 firer: LosLevel, 178 target: LosLevel,</pre>
omdurman-rules/src/los_table.rs:330 GH:omdurman-	has_los	<pre>328 /// `unit_level_at` closure returns the LOS level of 329 blocking units 330 /// (non-gunboat, non-fort per note a) in an intervening hex, or `None`. 331 pub fn has_los(</pre>

rules/src/ los_table.rs:330		331 board: &crate::board::BoardInfo, 332 from: HexCoord,
omdurman- types/src/ lib.rs:219 GH:omdurman- types/src/lib.rs:219	HexsideKind::blocks_los	217 /// (LOS table conditions 2-4, 7) and note (e) are 218 handled by the engine 219 /// in `omdurman_rules::los_table`, not by this 220 predicate. 221 pub fn blocks_los(self) -> bool { 222 matches!(self, HexsideKind::Wall 223 HexsideKind::Crest) 224 }
omdurman- types/src/ lib.rs:219 GH:omdurman- types/src/lib.rs:219	Terrain::blocks_los	217 /// (LOS table conditions 2-4, 7) and note (e) are 218 handled by the engine 219 /// in `omdurman_rules::los_table`, not by this 220 predicate. 221 pub fn blocks_los(self) -> bool { 222 matches!(self, HexsideKind::Wall 223 HexsideKind::Crest) 224 }
omdurman- types/src/ lib.rs:439 GH:omdurman- types/src/lib.rs:439	Terrain::is_los_trees	437 /// (§6.3 note 1). Retained for compatibility; the 438 full LOS engine 439 checks `Terrain::Trees` directly. 440 pub fn is_los_trees(self) -> bool { 441 matches!(self, Terrain::Trees { .. }) 442 }

Covered by tests: `los_level_mapping` `blocking_rules_all_cells_covered` `has_los_empty_board_is_clear`
`has_los_adjacent_clear` `has_los_howitzer_bypasses` `has_los_wall_hexside_blocks_ground_to_ground` `has_los_gate_hexside_passes`
`has_los_breach_hexside_passes` `has_los_rough_intervening_blocks_ground_to_ground`
`has_los_two_tree_hexes_pass_ground_to_ground` `has_los_three_tree_hexes_block_ground_to_ground`
`has_los_two_hut_hexes_pass_ground_to_ground` `has_los_three_hut_hexes_block_ground_to_ground`
`has_los_hilltop_to_hilltop_clear_no_units` `has_los_hilltop_to_hilltop_blocked_by_hilltop_unit`
`has_los_hilltop_to_hilltop_not_blocked_by_ground_unit` `has_los_rough_to_rough_unit_at_lower_level_passes`
`has_los_rough_to_rough_unit_at_same_level_blocks` `has_los_rough_to_rough_hilltop_blocks`
`has_los_ground_to_hilltop_intervening_hilltop_blocks` `has_los_building_blocks_like_huts_ground_to_ground`
`has_los_two_building_hexes_pass_ground_to_ground` `los_level_for_unit_gunboat_is_rough` `los_level_for_unit_fort_is_ground`
`los_level_for_unit_walled_city_adj_wall_is_rough` `gunboat_firer_uses_rough_row_not_ground` `los_reflexive_all_terrains`
`los_reflexive_hilltop` `los_symmetric_ground_to_ground_no_units` `los_howitzer_always_has_los` `los_howitzer_same_hex`
`los_blocking_rules_match_reference_table` `los_ground_to_ground_features_block_as_expected`

§6.6 – Special Artillery Capabilities

implemented

manual page unknown

Special Artillery Capabilities

File	Symbol	Code Snippet
omdurman- rules/src/ lib.rs:497 GH:omdurman- rules/src/lib.rs:497	WeaponClass	495 /// enum so a "spear" unit cannot accidentally fire on 496 the "Howitzer" line. 497 #[derive(Serialize, Deserialize, Clone, Copy, 498 PartialEq, Eq, Hash, Debug, strum::Display)] 499 pub enum WeaponClass { 500 /// Dervish spears and swords -- no ranged fire at 501 all. 502 Melee,

§6.7 – Defensive Fire

implemented

manual page unknown

Defensive Fire

In Defensive Fire phase, all of the non-moving player's units may fire at any of the moving player's units in range, within the limitations imposed by the rules of combat (see 6.1 to 6.6). There is no advance after combat as a result of defensive fires.

See also: §6.6

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:4015 GH:omdurman-rules/src/effects.rs:4015	can_advance_after_combat	<pre>4013 /// player's unit, not artillery, adjacent to `to`, and `to` now empty. 4014 /// Wall/khor hexside restrictions are not enforced (no hexside map data). 4015 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 4016 let unit = self.unit_or_err(unit_id)?; 4017 // §6.7: there is no advance after combat as a result of defensive fire.</pre>

Covered by tests: [no_advance_after_defensive_fire](#) [defensive_fire_opens_no_advance_window](#)

§6.11 – Fire combat factor printed on units

implemented

manual page unknown

The fire combat factor of the various unit types is printed directly on the units and is a numerical expression of the unit's fire strength.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:78 GH:omdurman-rules/src/lib.rs:78	FireFactor	<pre>76 /// Every possible value from the annotated counter set is a named variant. 77 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 78 pub enum FireFactor { 79 One = 1, 80 Three = 3,</pre>

Covered by tests: [fire_factor_sum_to_row](#)

§6.12 – Fire combat is always voluntary

implicit

manual page unknown

Fire combat is always voluntary. A unit is never required to fire at enemy units merely because they are in range or adjacent.

§6.13 – Fire factor is unitary (may not be divided)

implicit

manual page unknown

If a unit elects to fire, its fire combat factor at an enemy unit, that fire combat factor is unitary. A unit's fire combat factor may not be divided up to fire at enemy units on different hexes.

§6.14 – Players may combine fire factors into one attack

implemented

manual page unknown

Players may combine fire during fire combat phase, i.e. they may fire at an enemy-occupied hex with as many friendly units as may legally do so, combining all of their fire combat factors into one attack. Note that in any given fire combat phase, however, a combat unit may only fire once and may only be fired at once (exceptions: Maxim guns and gunboats — see 6.4).

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:92 GH:omdurman-rules/src/lib.rs:92	sum_to_row	<pre>90 impl FireFactor { 91 /// Sum multiple fire factors and return the corresponding Combat Results Table row (rulebook §6.11). 92 pub fn sum_to_row<'a>(factors: impl IntoIterator<Item = &'a FireFactor>) -> FireFactorRow { 93 let total: u16 = factors.into_iter().map(f f.value()).sum(); 94 crate::combat_results_table::FireFactorRow::from_total(total)</pre>

Covered by tests: [unit_may_only_be_fired_at_once_per_phase](#) [gunboat_and_maxim_may_be_fired_at_repeatedly](#)

§6.15 – May divide a stack to fire at different hexes

implicit

manual page unknown

Players may also divide a stack of units in order to fire at different enemy-occupied hexes. Anglo-Egyptian infantry units having brigade integrity, however, do not receive their +1 direct fire modifier unless they all fire at the same enemy-occupied hex (see 5.54).

See also: §5.54

§6.16 – Halving fire strength rounds down, minimum 1

implemented

manual page unknown

When halving fire combat strength, always round down each individual unit. For example, an Egyptian brigade of four battalions, each having a printed strength of 9 fire factors, will fire a total of 16 factors when halved. However, a unit's firing strength is never reduced below one by halving.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:532 GH:omdurman-rules/src/lib.rs:532	RangeBand	<pre>530 /// multiplied at a given distance (§6.22). 531 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 532 pub enum RangeBand { 533 Tripled, 534 Doubled,</pre>

§6.21 – First check LOS before firing

implemented

manual page unknown

When combat units wish to fire at enemy units, first check the Line of Sight Table to be sure the firing unit can see the target hex (exception: howitzer fire, see 6.64).

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:175 GH:omdurman-rules/src/los_table.rs:175	blocking_rules	<pre>173 /// if ALL conditions are satisfied (AND semantics). An 174 /// means the feature always blocks. 175 pub fn blocking_rules(176 firer: LosLevel, 177 target: LosLevel,</pre>
omdurman-rules/src/los_table.rs:330 GH:omdurman-rules/src/los_table.rs:330	has_los	<pre>328 /// `unit_level_at` closure returns the LOS level of 329 /// (non-gunboat, non-fort per note a) in an 330 /// intervening hex, or `None`. 331 pub fn has_los(332 board: &crate::board::BoardInfo,</pre>

Covered by tests: [can_fire_at_rejects_blocked_los](#) [can_fire_at_allows_clear_los](#)

§6.22 – Consult Range Effects Table

implemented

manual page unknown

Next consult the Range Effects Table to see if the firing unit's fire combat factor is tripled, doubled, normal, halved, or if the target hex is out of range. Add up the total number of fire combat factors firing at the enemy-occupied hex.

File	Symbol	Code Snippet
omdurman-rules/src/range_effects.rs:23 GH:omdurman-rules/src/range_effects.rs:23	ae_range_effects	<pre>21 /// Look up the range band for an Anglo-Egyptian weapon 22 /// (\$6.22, §6.24). 23 /// Distances > 10 are out of range for all weapons. 24 pub fn ae_range_effects(weapon: WeaponClass, distance: 25 HexDistance) -> RangeBand { 26 if distance.value() > 10 { 27 return RangeBand::OutOfRange;</pre>
omdurman-rules/src/range_effects.rs:61 GH:omdurman-rules/src/range_effects.rs:61	dervish_range_effects	<pre>59 /// Look up the range band for a Dervish weapon 60 /// (\$6.22). 61 /// Distances > 10 are out of range for all weapons. 62 pub fn dervish_range_effects(weapon: WeaponClass, 63 distance: HexDistance) -> RangeBand { 64 if distance.value() > 10 { 65 return RangeBand::OutOfRange;</pre>
omdurman-rules/src/lib.rs:516 GH:omdurman-rules/src/lib.rs:516	Range	<pre>514 /// (rulebook §6.22). Distances beyond 10 hexes are out 515 /// of range for all weapons. 516 #[derive(Clone, Copy, PartialEq, Eq, Debug)] 517 pub enum Range { 518 One, 519 Two,</pre>
omdurman-rules/src/lib.rs:532 GH:omdurman-rules/src/lib.rs:532	RangeBand	<pre>530 /// multiplied at a given distance (§6.22). 531 #[derive(Serialize, Deserialize, Clone, Copy, 532 PartialEq, Eq, Debug)] 533 pub enum RangeBand { 534 Tripled, 535 Doubled,</pre>
omdurman-rules/src/lib.rs:179 GH:omdurman-rules/src/lib.rs:179	HexDistance	<pre>177 /// (rulebook §6.22, §7.5). 178 #[derive(Serialize, Deserialize, Clone, Copy, 179 PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 180 pub struct HexDistance(u16); 181 impl HexDistance {</pre>

Covered by tests: `ae_rifles_doubled_at_range_1` `ae_rifles_halved_at_range_4` `ae_howitzer_range` `dervish_rifles_shorter_range` `melee_only_range_1` `ae_range_effects_artillery_full` `ae_range_effects_maxims_match_rifles` `ae_range_effects_distance_over_10` `ae_range_effects_howitzer_halved_4_to_10` `dervish_range_effects_rifles` `dervish_range_effects_artillery` `dervish_range_effects_maxims_and_howitzer` `dervish_range_effects_melee` `dervish_range_effects_distance_over_10` `fire_combat_eliminate_target` `max_day_range_all_combos` `range_effects_every_cell_ae_artillery` `range_effects_every_cell_ae_howitzer` `range_effects_every_cell_ae_maxims` `range_effects_every_cell_ae_rifles` `range_effects_every_cell_dervish_artillery` `range_effects_every_cell_dervish_maxims_howitzer` `range_effects_every_cell_dervish_rifles` `range_effects_every_cell_dervish_spears` `can_fire_at_gates_phase_range_and_player` `mixed_attack_bands_per_firer` `ae_range_effects_monotone_non_increasing` `ae_howitzer_has_minimum_range` `dervish_range_effects_monotone_non_increasing` `range_effects_first_range_max_effect_last_range_oor`

§6.23 – Terrain defensive modifier

implemented

manual page unknown

Next check the Terrain Effects Chart to see if the enemy-occupied hex fired upon contains any terrain which gives the enemy units in that hex a defensive benefit. If so, apply this negative modifier to the roll of the ten-sided die and cross-index your net die roll on the Combat Results Table with the number of combat factors firing.

File	Symbol	Code Snippet
omdurman-rules/src/terrain_chart.rs:59 GH:omdurman-rules/src/terrain_chart.rs:59	defense_modifier	<pre> 57 58 /// Convenience: get the defense modifier for a terrain type (rulebook §6.23, Terrain Effects Chart). 59 pub fn defense_modifier(terrain: Terrain) -> i16 { 60 terrain_effects_chart(terrain).defense_modifier 61 }</pre>
omdurman-rules/src/lib.rs:923 GH:omdurman-rules/src/lib.rs:923	FireModifier::Terrain	<pre> 921 /// Negative modifier from the Terrain Effects Chart applied to the 922 /// defender's hex (§6.23). 923 Terrain(i16), 924 /// -2 thorn-hedge defensive modifier (§9.231). 925 ZaribaThornHedge,</pre>

Covered by tests: `clear_terrain_no_bonus` `building_gives_minus_3` `palm_grove_gives_minus_2` `rough_movement_and_defense` `swamp_movement_and_defense` `hilltop_movement_and_defense` `huts_movement_and_defense` `defense_modifier_convenience_matches_chart` `terrain_movement_costs_in_bounds` `terrain_defense_modifier_non_positive`

§6.24 – Anglo-Egyptian direct fire accuracy bonus and brigade integrity

implemented

manual page unknown

All Anglo-Egyptian direct fire attacks receive a +1 modifier to their die roll as an accuracy bonus. In addition, any stack of Anglo-Egyptian infantry having brigade integrity (see 5.54) receives a +1 modifier to their die roll if all four fire at the same enemy-occupied hex. These modifiers are cumulative.

See also: §5.54

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:917 GH:omdurman-rules/src/lib.rs:917	AngloEgyptianDirectFire	<pre> 915 pub enum FireModifier { 916 /// +1 to all Anglo-Egyptian *direct* fire (§6.24). 917 AngloEgyptianDirectFire, 918 /// +1 brigade integrity, applied only if all four battalions fire at 919 /// the same enemy-occupied hex (§5.54, §6.24).</pre>
omdurman-rules/src/lib.rs:902	BrigadeIntegrity	<pre> 900 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 901 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]</pre>

GH:omdurman-rules/src/lib.rs:902		<pre> 902 pub enum BrigadeIntegrity { 903 None, 904 Integrated(BrigadeId), </pre>
omdurman-rules/src/lib.rs:933 GH:omdurman-rules/src/lib.rs:933	FireModifier::die_modifier	<pre> 931 impl FireModifier { 932 /// Return the numeric die-roll modifier for this bonus/penalty (rulebook \$6.24, \$5.54, \$6.23, \$9.231, \$9.232). 933 pub fn die_modifier(self) -> i16 { 934 match self { 935 FireModifier::AngloEgyptianDirectFire FireModifier::BrigadeIntegrity => 1, </pre>

Covered by tests: [brigade_integrity_modifier_is_engine_derived](#) [fire_modifiers_are_engine_derived_and_mismatches_rejected](#)

§6.41 – Direct Fire Subphase

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:327 GH:omdurman-rules/src/lib.rs:327	DirectFire	<pre> 325 pub enum FireSubPhase { 326 /// Direct fire (§6.41). Both sides participate in this sub-phase. 327 DirectFire, 328 /// Anglo-Egyptian only: Maxim second fire + named- gunboat howitzer fire (§6.42). 329 MaximSecondAndHowitzer, </pre>

§6.42 – Maxim Second Fire and Howitzer Fire Subphase

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:329 GH:omdurman-rules/src/lib.rs:329	MaximSecondAndHowitzer	<pre> 327 DirectFire, 328 /// Anglo-Egyptian only: Maxim second fire + named- gunboat howitzer fire (§6.42). 329 MaximSecondAndHowitzer, 330 } 331 </pre>
omdurman-types/src/lib.rs:912 GH:omdurman-types/src/lib.rs:912	fires_twice	<pre> 910 911 /// Maxim guns fire twice per turn -- once in the Direct Fire Subphase and 912 /// again in the Maxim Second Fire Subphase (rulebook \$6.42). 913 pub fn fires_twice(self) -> bool { 914 matches!(self, UnitKind::Maxim { .. }) </pre>

Covered by tests: [howitzer_on_target_7_to_10](#) [howitzer_scatters_below_7](#) [howitzer_short_on_5_6](#) [howitzer_long_on_3_4](#) [howitzer_left_right_on_1_2](#) [advance_window_bridges_fire_subphase_and_closes_at_melee](#) [fired_at_tracker_resets_at_maxim_subphase](#)

§6.51 – Leader Units

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:440	BritishLeader	<pre> 438 /// to claim the Mahdi's Tomb (§9.14). 439 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] </pre>

GH:omdurman-rules/src/lib.rs:440		<pre> 440 pub enum BritishLeader { 441 Kitchener, 442 Gatacre, </pre>
omdurman-types/src/lib.rs:865 GH:omdurman-types/src/lib.rs:865	BritishLeader	<pre> 863 /// Dervish leader (§6.51): fire / melee / movement. May melee attack (§7.4). 864 DervishLeader { fire: i32, melee: i32, movement: i32 }, 865 /// Anglo-Egyptian leader (§6.51): movement only. 866 BritishLeader { movement: i32 }, 867 /// Wall-breach marker placed by artillery fire (\$6.63). Not a combat unit. </pre>
omdurman-types/src/lib.rs:906 GH:omdurman-types/src/lib.rs:906	has_combat_factors	<pre> 904 905 /// British leaders print a movement factor only (\$6.51); other kinds carry 906 /// fire and/or melee factors. Markers carry no stats. 907 pub fn has_combat_factors(self) -> bool { 908 !matches!(self, UnitKind::BritishLeader { .. } UnitKind::Marker UnitKind::Breach UnitKind::BareCounter) </pre>

Covered by tests: `zero_factor_is_none_not_zero` `kitchener_block_resolves_leaders_friendlys_camel_and_sudanese` `dervish_leader_sections_resolve_leader_and_retinue_per_cell`

§6.52 – Anglo-Egyptian Friendlys Brigade

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:631 GH:omdurman-rules/src/lib.rs:631	is_friendlys	<pre> 629 /// "Friendlys" units obey several special rules (\$5.21, §5.23, §6.52, 630 /// §9.14 victory conditions). 631 pub fn is_friendlys(&self) -> bool { 632 matches!(633 self, </pre>
omdurman-types/src/lib.rs:931 GH:omdurman-types/src/lib.rs:931	Friendlys	<pre> 929 Sudanese, 930 /// Native volunteer brigade -- the Shaggyeh (\$6.52). Do not receive 931 /// brigade integrity (§5.54 enumerates only British/Egyptian/Sudanese). 932 Friendlys, 933 } </pre>

Covered by tests: `friendlys_counters_score_by_bank_not_as_leaders` `friendlys_validate_and_resolve_on_dervish_table`

§6.53 – Royal Engineers demolition

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:600 GH:omdurman-rules/src/lib.rs:600	RoyalEngineers	<pre> 598 /// The Royal Engineers (§6.53) -- a *specific* unit, not a class, so we 599 /// model it explicitly. 600 RoyalEngineers, 601 } 602 </pre>
omdurman-rules/src/lib.rs:818	demolishing	<pre> 816 /// Set when the Royal Engineers are committed to a demolition this turn 817 /// (§6.53) -- neither offensive fire nor melee </pre>

GH:omdurman-rules/src/lib.rs:818		<pre> allowed that turn. 818 pub demolishing: bool, 819 /// Set when a gunboat has lost its engines to a river mine (§10.12, roll 820 /// 5-7): it may no longer move under power and instead drifts two hexes per </pre>
omdurman-rules/src/effects.rs:211 GH:omdurman-rules/src/effects.rs:211	Demolition	<pre> 209 210 /// Royal Engineers demolition (rulebook §6.53). 211 Demolition { 212 unit_id: UnitId, 213 target: DemolitionTarget, </pre>
omdurman-rules/src/effects.rs:4437 GH:omdurman-rules/src/effects.rs:4437	apply_demolition	<pre> 4435 /// resolution happens at end of turn via [`apply_resolve_demolition`], which 4436 /// checks the engineer is still adjacent and undisrupted. 4437 pub fn apply_demolition(4438 state: &mut GameState, 4439 unit_id: UnitId, </pre>
omdurman-rules/src/lib.rs:1058 GH:omdurman-rules/src/lib.rs:1058	DemolitionTarget	<pre> 1056 /// disrupted or driven off. 1057 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1058 pub enum DemolitionTarget { 1059 Fort(UnitId), 1060 WallHexside(HexsideRef), </pre>

Covered by tests: `demolition_cancelled_when_engineer_disrupted` `demolition_cancelled_when_engineer_moved_away`

§6.54 – Forts

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:864 GH:omdurman-rules/src/lib.rs:864	ZocReason	<pre> 862 /// Used by the engine when answering "is this hex in an enemy ZOC?". 863 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 864 pub enum ZocReason { 865 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 866 /// leader (§5.41) projects ZOC into each of its six adjacent hexes. </pre>
omdurman-rules/src/lib.rs:871 GH:omdurman-rules/src/lib.rs:871	Fort	<pre> 869 GunboatVsGunboat, 870 /// Forts project ZOC out of, but not into, an empty fort (§5.44, §6.54). 871 Fort, 872 /// Walled-city ZOC: extends out through walls and gates but not in, 873 /// across a breach in both directions (§5.44). </pre>
omdurman-rules/src/lib.rs:828 GH:omdurman-rules/src/lib.rs:828	UnitState::may_act	<pre> 826 impl UnitState { 827 /// A disrupted unit may not move, fire, or melee (rulebook §5, reference notes). 828 pub fn may_act(self) -> bool { 829 !self.disrupted 830 } </pre>
omdurman-rules/src/lib.rs:806 GH:omdurman-rules/src/lib.rs:806	UnitState	<pre> 804 /// rather than one big enum. 805 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, Default)] 806 pub struct UnitState { 807 /// Reference table: "Disrupted units: no ZOC; may not move; may not fire </pre>

		808 /// offensively or defensively; may not melee; are turned face up at the
omdurman-rules/src/lib.rs:962 GH:omdurman-rules/src/lib.rs:962	FireAttack	960 /// modifiers (rulebook §6). 961 #[derive(Serialize, Deserialize, Clone, Debug)] 962 pub struct FireAttack { 963 pub firing_player: Player, 964 pub phase: Phase,
omdurman-rules/src/lib.rs:977 GH:omdurman-rules/src/lib.rs:977	FireAttack::net_modifier	975 impl FireAttack { 976 /// Sum of all fire modifiers applied to this attack (rulebook §6.24). 977 pub fn net_modifier(&self) -> i16 { 978 self.modifiers.iter().map(m m.die_modifier()).sum() 979 }

Covered by tests: [retreat_before_melee_may_not_land_on_enemy_fort](#)

§6.61 – Only artillery may fire at gunboats; 3+ to sink

implemented

manual page unknown

Only artillery may fire at gunboats. A result of 3 or more on the combat results table is required to sink a gunboat. Any other result is a miss.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:497 GH:omdurman-rules/src/lib.rs:497	WeaponClass	495 /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 496 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 497 pub enum WeaponClass { 498 /// Dervish spears and swords -- no ranged fire at all. 499 Melee,

Covered by tests: [rifles_may_not_sink_a_gunboat](#)

§6.62 – Only artillery may fire at forts; 2+ to destroy

implemented

manual page unknown

Only artillery may fire at forts. A result of 2 or more on the combat results table is required to eliminate a fort. Any other result is a miss. If the fort contains any enemy units at the instant it is destroyed, one unit is eliminated with the fort.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:497 GH:omdurman-rules/src/lib.rs:497	WeaponClass	495 /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 496 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 497 pub enum WeaponClass { 498 /// Dervish spears and swords -- no ranged fire at all. 499 Melee,

§6.63 – Only artillery may breach wall hexsides; 2+ required

implemented

manual page unknown

Only artillery may fire to breach a wall hexside of Khartoum or the walled city of Omdurman. A result of 2 or more on the combat results table is required to breach a wall. Any other result is a miss. The effect of the breach is to negate the wall hexside for line of sight purposes. Place a “BREACH” marker in an adjacent hex so that the arrow points to the breached hexside. If any enemy units are adjacent to the wall hexside at the instant it is breached, one enemy unit is eliminated.

****6.64) Howitzer fire:**** Five units in the game have howitzer fire capability. These are the five named British gunboats. They may fire their artillery factor as direct fire during the Direct Fire Subphase (see 4 and 6.41) and may then fire the same artillery factor as howitzer fire during the Maxim Second Fire and Howitzer Subphase (see 4 and 6.42). Exception: no howitzer fire is allowed during night game turns. To fire howitzer fire, select any target hex between 4 and 10 hexes from the firing gunboat (ignoring the Line of Sight Table) and roll the ten-sided die twice. The first die roll is the Combat Results Table die roll and the second roll is the impact hex die roll. Refer to the Howitzer Fire Scattergram on the mapsheet for the impact hex. The designated target hex is hit on a roll of 7-10. (see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:187 GH:omdurman-types/src/lib.rs:187	Breach	<pre>185 /// Breach in a wall (artillery/§6.63 or Royal 186 Engineers/§6.53). ZOC both 187 /// ways; LOS no longer blocked across the hexside. 188 Breach, 189 /// Khor -- gully/wadi. ZOCs do not extend across 190 (§5.44); advance after 191 /// combat may not cross (§6.82).</pre>
omdurman-rules/src/effects.rs:307 GH:omdurman-rules/src/effects.rs:307	ArtilleryBreachWall	<pre>305 /// pre-rolled d10 used for the CRT lookup; range/ 306 LOS are re-derived by the 307 /// engine from the firers and `target`. 308 ArtilleryBreachWall { 309 firers: Vec<UnitId>, 310 target: HexsideRef,</pre>
omdurman-rules/src/effects.rs:4590 GH:omdurman-rules/src/effects.rs:4590	apply_artillery_breach_v	<pre>4588 /// artillery's CRT roll -- the rulebook specifies the 4589 same "2+ required" 4590 /// threshold for both trigger styles. 4591 pub fn apply_artillery_breach_wall(4592 state: &mut GameState, 4593 firers: &[UnitId],</pre>
omdurman-rules/src/effects.rs:1996 GH:omdurman-rules/src/effects.rs:1996	can_fire_at_wall	<pre>1994 /// range band and resolving the CRT – this method 1995 only validates one 1996 /// firer at a time. 1997 pub fn can_fire_at_wall(1998 &self, 1999 firer: UnitId,</pre>

Covered by tests: [breach_marker_cell_returns_none](#)

§6.64 – Howitzer fire

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/howitzer_scatter.rs:6 GH:omdurman-rules/src/howitzer_scatter.rs:6	ScatterDirection	<pre>4 /// (§6.64). The caller maps these to hex-grid offsets. 5 #[derive(Clone, Copy, PartialEq, Eq, Debug)] 6 pub enum ScatterDirection { 7 /// Roll 7-10: hit the target hex. 8 OnTarget,</pre>

omdurman-rules/src/howitzer_scatter.rs:28 GH:omdurman-rules/src/howitzer_scatter.rs:28	howitzer_scatter	<pre> 26 /// 3-4 [`ScatterDirection::Long`] (upstream) 27 /// 1-2 [`ScatterDirection::LeftRight`] 28 pub fn howitzer_scatter(impact_roll: DieRoll) -> ScatterDirection { 29 use DieRoll::*; 30 match impact_roll { </pre>
omdurman-rules/src/lib.rs:451 GH:omdurman-rules/src/lib.rs:451	GunboatId	<pre> 449 /// fire; "old" gunboats do not (rulebook §2.32). 450 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 451 pub enum GunboatId { 452 /// One of the five new-type named gunboats with howitzer capability. 453 Named(NamedGunboat), </pre>
omdurman-rules/src/effects.rs:124 GH:omdurman-rules/src/effects.rs:124	HowitzerFire	<pre> 122 /// - CRT result applied to units at impact hex (not the original target). 123 /// - Firers marked as fired. 124 HowitzerFire { 125 attack: FireAttack, 126 combat_results_table_roll: DieRoll, </pre>
omdurman-rules/src/effects.rs:3319 GH:omdurman-rules/src/effects.rs:3319	apply_howitzer_fire	<pre> 3317 3318 /// Validate and apply a howitzer fire attack (scatter path) (rulebook §6.64). 3319 pub fn apply_howitzer_fire(3320 state: &mut GameState, 3321 attack: &FireAttack, </pre>
omdurman-rules/src/lib.rs:998 GH:omdurman-rules/src/lib.rs:998	HowitzerResolution	<pre> 996 /// roll on the Howitzer Fire Scattergram (§6.64). 997 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 998 pub struct HowitzerResolution { 999 pub combat_results_table_roll: DieRoll, 1000 pub impact_roll: DieRoll, </pre>
omdurman-rules/src/lib.rs:1005 GH:omdurman-rules/src/lib.rs:1005	HowitzerResolution::hit_target_hex	<pre> 1003 impl HowitzerResolution { 1004 /// The designated target hex is hit on impact roll 7-10 (§6.64). 1005 pub fn hit_target_hex(self) -> bool { 1006 use DieRoll::*; 1007 matches!(self.impact_roll, Seven Eight Nine Ten) </pre>
omdurman-rules/src/effects.rs:1783 GH:omdurman-rules/src/effects.rs:1783	can_fire_at	<pre> 1781 /// modifier in the [`FireAttack`] and is responsible for the LOS gate. 1782 /// (Howitzer fire ignores LOS entirely -- §6.64.) 1783 pub fn can_fire_at(1784 &self, 1785 firer: UnitId, </pre>

Covered by tests: [howitzer_on_target_7_to_10](#) [howitzer_scatters_below_7](#) [howitzer_short_on_5_6](#) [howitzer_long_on_3_4](#) [howitzer_left_right_on_1_2](#) [named_and_old_gunboats_resolve](#) [named_gunboat_has_howitzer](#) [named_gunboat_may_fire_howitzer_in_second_subphase](#) [named_gunboat_direct_fire_uses_artillery_weapon](#) [named_gunboat_no_howitzer_at_night](#) [dervish_gunboat_lacks_howitzer](#)

§6.81 – Moving player may fire with all capable units

implicit

manual page unknown

During Offensive Fire phase, the moving player may fire with all of his units capable of firing, up to their maximum range, within the limitations imposed by the rules of combat.

§6.82 – Advance after combat (offensive fire)

implemented

manual page unknown

If an enemy-occupied hex is vacated as a result of offensive fire, friendly units may advance after combat into the vacated hex. To be eligible to advance, the friendly units must have participated in the attack and must have been adjacent to the vacated hex. Note that artillery may not advance, nor may units advance across a wall hexside (except at a gate or breach). Units may never advance after combat across a khor.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:198 GH:omdurman-rules/src/effects.rs:198	AdvanceAfterCombat	<pre> 196 /// **Postconditions:** Unit position moved to `to`, `vacated_by_combat` 197 /// entry consumed. 198 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 199 200 // -- Unit state changes ----- </pre>
omdurman-rules/src/effects.rs:4335 GH:omdurman-rules/src/effects.rs:4335	apply_advance_after_combat	<pre> 4333 4334 /// Apply an advance-after-combat for a unit (rulebook §6.82, §7.6). 4335 pub fn apply_advance_after_combat(4336 state: &mut GameState, 4337 unit_id: UnitId, </pre>
omdurman-types/src/lib.rs:230 GH:omdurman-types/src/lib.rs:230	blocks_advance_after_combat	<pre> 228 229 /// Whether advance-after-combat may *not* cross this side (§6.82, §7.6). 230 pub fn blocks_advance_after_combat(self) -> bool { 231 matches!(self, 232 Self:: </pre>
omdurman-rules/src/effects.rs:4015 GH:omdurman-rules/src/effects.rs:4015	can_advance_after_combat	<pre> 4013 /// player's unit, not artillery, adjacent to `to`, and `to` now empty. 4014 /// Wall/khor hexside restrictions are not enforced (no hexside map data). 4015 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 4016 let unit = self.unit_or_err(unit_id)?; 4017 // §6.7: there is no advance after combat as a result of defensive fire. </pre>

Covered by tests: [can_advance_after_combat_rejects_wall_hexside](#) [can_advance_after_combat_rejects_khor_hexside](#) [advance_requires_combat_vacated_hex](#) [advance_requires_participation](#)

8/8 implemented

§7 – Melee Phase

§7 – Melee Phase (chapter)

implemented

manual page unknown

Melee Phase

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:934 GH:omdurman-rules/src/effects.rs:934	PendingMelee	<pre>932 /// resolution after the reaction window is deterministic and host-ordered (rulebook §7.5). 933 #[derive(Serialize, Deserialize, Clone, Debug)] 934 pub struct PendingMelee { pub attack: MeleeAttack, 935 pub attacker_roll: DieRoll,</pre>

Covered by tests: [declared_melee_blocks_phase_advance](#)

§7.1 – Melee strength printed on counter

implemented

manual page unknown

The melee strength of all units is printed on the counter. Note that gunboats have no melee strength. Gunboats may neither melee attack nor be melee attacked.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:102 GH:omdurman-rules/src/lib.rs:102	MeleeFactor	<pre>100 /// Every possible value from the annotated counter set is a named variant. 101 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 102 pub enum MeleeFactor { One = 1, 103 Three = 3,</pre>
omdurman-rules/src/lib.rs:113 GH:omdurman-rules/src/lib.rs:113	MeleeFactor::sum	<pre>111 impl MeleeFactor { 112 /// Sum multiple melee factors (rulebook §7.1). 113 pub fn sum<'a>(factors: impl IntoIterator<Item = &'a MeleeFactor>) -> u16 { 114 factors.into_iter().map(f f.value()).sum() 115 }</pre>
omdurman-types/src/lib.rs:889 GH:omdurman-types/src/lib.rs:889	may_be_melee_attacked	<pre>887 } 888 889 /// Gunboats neither attack nor are attacked in melee (§7.1). 890 pub fn may_be_melee_attacked(self) -> bool { 891 !matches!(self, UnitKind::Gunboat { .. })</pre>

Covered by tests: [reinforcement_rejected_onto_enemy_occupied_hex](#)

§7.2 – Melee adjacent only, not across wall hexsides

implemented

manual page unknown

Melee simulates the hand-to-hand fighting of the period. Units may melee attack adjacent enemy units only. Units may not melee attack across a wall hexside, but may melee attack through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:225 GH:omdurman-types/src/lib.rs:225	blocks_melee	<pre> 223 /// Whether melee may *not* be made across this side (§7.2). Gates and 224 /// breaches are passable to melee. pub fn blocks_melee(self) -> bool { 225 matches!(self, HexsideKind::Wall 226 HexsideKind::ZaribaThornHedge) } 227 </pre>
omdurman-rules/src/effects.rs:2107 GH:omdurman-rules/src/effects.rs:2107	can_melee	<pre> 2105 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 2106 /// thorn-hedge hexside blocks the attack (§7.2). pub fn can_melee(&self, attacker: UnitId, 2107 defender_hex: HexCoord) -> Result<(), RuleError> { 2108 let unit = self.unit_or_err(attacker)?; 2109 </pre>

Covered by tests: [can_melee_gates_phase_adjacency_and_kind](#) [can_melee_rejects_wall_hexside](#) [can_melee_rejects_thorn_hedge_hexside](#) [can_melee_allows_gate_hexside](#)

§7.3 – Simultaneous melee combat

implemented

manual page unknown

Melee combat is considered simultaneous, so that units eliminated by melee attacks still get a melee combat die roll.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:147 GH:omdurman-rules/src/effects.rs:147	MeleeCombat	<pre> 145 /// - Winner may advance into vacated hex (§7.6). 146 /// - Victory points awarded for eliminations. 147 MeleeCombat { 148 attack: MeleeAttack, 149 attacker_roll: DieRoll, </pre>
omdurman-rules/src/effects.rs:3688 GH:omdurman-rules/src/effects.rs:3688	apply_melee_combat	<pre> 3686 3687 /// Apply a simultaneous melee combat between two adjacent hexes (rulebook §7). 3688 pub fn apply_melee_combat(3689 state: &mut GameState, 3690 attack: &MeleeAttack, </pre>

§7.4 – Who may melee attack / defend

implemented

manual page unknown

Only infantry, cavalry, camel units, and Dervish leaders may melee attack. All units (except gunboats — see 7.1) may melee defend.

See also: §7.1

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:878 GH:omdurman-types/src/lib.rs:878	may_melee_attack	<pre> 876 impl UnitKind { 877 /// Rulebook §7.4 -- only infantry, cavalry, camel and Dervish leaders may 878 /// melee *attack*. All others (except gunboats) may melee *defend* (§7.1). 879 pub fn may_melee_attack(self) -> bool { 880 matches!(</pre>

omdurman-types/src/lib.rs:843 GH:omdurman-types/src/lib.rs:843	UnitKind	<pre> 841 /// (`Some(UnitKind::...)`) ; a non-unit marker counter carries 842 /// `Some(UnitKind::Marker)` or `None`. 843 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 844 pub enum UnitKind { 845 /// Foot infantry (§2.3): fire / melee / movement.</pre>
omdurman-types/src/lib.rs:694 GH:omdurman-types/src/lib.rs:694	DervishTribe	<pre> 692 #[derive(693 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display, strum::EnumIter, 694)] 695 pub enum DervishTribe { 696 Baggara,</pre>
omdurman-rules/src/effects.rs:2107 GH:omdurman-rules/src/effects.rs:2107	can_melee	<pre> 2105 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 2106 /// thorn-hedge hexside blocks the attack (§7.2). 2107 pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> { 2108 let unit = self.unit_or_err(attacker)?; 2109 </pre>

§7.5 – Cavalry/camel retreat before melee

implemented

manual page unknown

Cavalry and camel units may retreat two hexes from an infantry melee attack. Note, however, that only one retreat per unit per turn is permitted. Thus, if their retreat places them adjacent to enemy units whose melee attacks have not yet been resolved, those enemy units may elect to attack the retreating unit(s).

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:184 GH:omdurman-rules/src/effects.rs:184	RetreatBeforeMelee	<pre> 182 /// 183 /// **Postconditions:** Unit position moved to `to`. 184 RetreatBeforeMelee { unit_id: UnitId, to: HexCoord }, 185 186 /// An attacking unit advances into a hex vacated by combat (rulebook §6.82</pre>
omdurman-rules/src/effects.rs:4297 GH:omdurman-rules/src/effects.rs:4297	apply_retreat_before_melee	<pre> 4295 4296 /// Apply a retreat-before-melee for a cavalry/camel unit (rulebook §7.5). 4297 pub fn apply_retreat_before_melee(4298 state: &mut GameState, 4299 unit_id: UnitId,</pre>
omdurman-types/src/lib.rs:895 GH:omdurman-types/src/lib.rs:895	may_retreat_before_melee	<pre> 893 894 /// Cavalry and camel units may retreat two hexes from an infantry melee 895 /// attack (§7.5). 896 pub fn may_retreat_before_melee(self) -> bool { 897 matches!(self, UnitKind::Cavalry { .. } UnitKind::Camel { .. })</pre>
omdurman-rules/src/lib.rs:179 GH:omdurman-rules/src/lib.rs:179	HexDistance	<pre> 177 /// (rulebook §6.22, §7.5). 178 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 179 pub struct HexDistance(u16); 180 181 impl HexDistance {</pre>
omdurman-rules/src/effects.rs:3958 GH:omdurman-	can_retreat_before_melee	<pre> 3956 /// two hexes away and empty. (Does not verify the attacker is infantry -- 3957 /// the caller offers the retreat only in response to one.)</pre>

rules/src/ effects.rs:3958		<pre> 3958 pub fn can_retreat_before_melee(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 3959 let unit = self.unit_or_err(unit_id)?; 3960 if !matches!(self.phase, Phase::Melee) { </pre>
-------------------------------	--	---

Covered by tests: [retreat_before_melee_only_cavalry_two_hexes](#) [retreat_opens_window_only_when_hex_empties](#)

§7.6 – Advance after melee

implemented

manual page unknown

If a melee attack eliminates all of the defenders in an adjacent hex, the Dervish player **MUST** advance into the vacated hex. To be eligible to advance, the Dervish units must have been adjacent to the vacated hex and participated in the melee attack that eliminated the defenders. All surviving eligible Dervish units **MUST** advance, up to the stacking limit. The Anglo-Egyptian player may advance if desired. Note that only attacking units may advance.

File	Symbol	Code Snippet
omdurman- rules/src/ effects.rs:198 GH:omdurman- rules/src/ effects.rs:198	AdvanceAfterCombat	<pre> 196 /// **Postconditions:** Unit position moved to `to`; `vacated_by_combat` 197 /// entry consumed. 198 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 199 200 // -- Unit state changes ----- </pre>
omdurman- rules/src/ effects.rs:4335 GH:omdurman- rules/src/ effects.rs:4335	apply_advance_after_combat	<pre> 4333 4334 /// Apply an advance-after-combat for a unit (rulebook \$6.82, \$7.6). 4335 pub fn apply_advance_after_combat(4336 state: &mut GameState, 4337 unit_id: UnitId, </pre>
omdurman- rules/src/ effects.rs:4015 GH:omdurman- rules/src/ effects.rs:4015	can_advance_after_combat	<pre> 4013 /// player's unit, not artillery, adjacent to `to`, and `to` now empty. 4014 /// Wall/khor hexside restrictions are not enforced (no hexside map data). 4015 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 4016 let unit = self.unit_or_err(unit_id)?; 4017 // \$6.7: there is no advance after combat as a result of defensive fire. </pre>

§7.7 – Melee modifiers

implemented

manual page unknown

To resolve melee, both the attacker and the defender roll on the Combat Results Table and apply the applicable melee modifier to their die roll. The Dervish player receives a +2 melee modifier, the Anglo-Egyptian player receives a +1 melee modifier. No terrain modifiers are applied to melee combat (Exception: Zariba hexsides in the historical scenario and the campaign game, if constructed — see 9.23). Melee losses must be taken from meleeing units first!

See also: §9.23

File	Symbol	Code Snippet
omdurman- rules/src/ lib.rs:1016 GH:omdurman-	MeleeModifier	<pre> 1014 1015 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1016 pub enum MeleeModifier { </pre>

rules/src/ lib.rs:1016		<pre> 1017 /// +2 to all Dervish melee rolls (§7.7). 1018 DervishStandard,</pre>
omdurman-rules/src/lib.rs:225 GH:omdurman-rules/src/lib.rs:225	DieModifier	<pre> 223 /// A die-roll modifier from a single named source 224 (rulebook §6.24, §7.7). 225 #[derive(Serialize, Deserialize, Clone, Copy, 226 PartialEq, Eq, Debug, Default)] 227 pub enum DieModifier { 228 #[default] 229 Zero,</pre>
omdurman-rules/src/lib.rs:1039 GH:omdurman-rules/src/lib.rs:1039	MeleeAttack	<pre> 1037 /// A melee attack: simultaneous, both sides roll on 1038 the Combat Results Table (§7.3, §7.7). 1039 #[derive(Serialize, Deserialize, Clone, Debug)] 1040 pub struct MeleeAttack { 1041 pub attacker_player: Player,</pre>
omdurman-rules/src/lib.rs:1020 GH:omdurman-rules/src/lib.rs:1020	MeleeModifier::Anglo-EgyptianStandard	<pre> 1018 DervishStandard,</pre> <pre> 1019 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 1020 AngloEgyptianStandard,</pre> <pre> 1021 /// Inverted to -2 when Dervish units melee-attack 1022 across a trench into 1023 an entrenched defender (§9.232).</pre>
omdurman-rules/src/lib.rs:1023 GH:omdurman-rules/src/lib.rs:1023	MeleeModifier::DervishVsDefender	<pre> 1021 /// Inverted to -2 when Dervish units melee-attack 1022 across a trench into 1023 an entrenched defender (§9.232). 1024 DervishVsTrenchedDefender,</pre>
omdurman-rules/src/lib.rs:1018 GH:omdurman-rules/src/lib.rs:1018	MeleeModifier::DervishStandard	<pre> 1016 pub enum MeleeModifier { 1017 /// +2 to all Dervish melee rolls (§7.7). 1018 DervishStandard,</pre> <pre> 1019 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 1020 AngloEgyptianStandard,</pre>

Covered by tests: [melee_resolves_simultaneously](#) [melee_modifiers_are_engine_derived_and_mismatches_rejected](#)

■ 2/2 implemented

§8 – Night Game Turns

§8.1 – Night effects

implemented

manual page unknown

The effects of night game turns are: a) all Anglo-Egyptian movement allowances are halved (round down), b) there is no Anglo-Egyptian howitzer fire, and c) all fire ranges are halved for both sides (round down, but range 1 stays range 1). Range effects on fire combat are the same as during day game turns. For example, an Anglo-Egyptian infantry unit firing at night will be doubled at range 1, normal at range 2, and may not fire at range 3 or greater.

****8.2) Dervish Desertion Roll:**** Once each campaign game, during the first night turn of the game, the Dervish player rolls one die to see how many of his units desert. The roll is made during the movement phase and the number of deserting Dervish units is equal to 1½ times the roll of one die. The Dervish player may choose which units desert by merely removing them from the mapsheet. The KHALIFA unit, gunboats, artillery units, and forts are the only Dervish units that may not be chosen. No victory points are awarded to the Anglo-Egyptian player for deserting Dervishes.

(see manual for full text)

See also: §8.2

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:148 GH:omdurman-rules/src/lib.rs:148	MovementAllowance::halve	<pre>146 impl MovementAllowance { 147 /// Night movement allowance = halved (round down) (rulebook §8.1, §5.11). 148 pub fn halve(self) -> Self { 149 let v = self.value() / 2; 150 MovementAllowance::try_from(v).expect("halved value always a named variant")</pre>
omdurman-types/src/lib.rs:757 GH:omdurman-types/src/lib.rs:757	DayNight	<pre>755 /// Anglo-Egyptian movement and all fire ranges, and forbid howitzer fire 756 /// (rulebook §8.1). 757 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 758 pub enum DayNight { 759 Day,</pre>
omdurman-rules/src/range_effects.rs:116 GH:omdurman-rules/src/range_effects.rs:116	night_max_range	<pre>114 115 /// The halved maximum range at night (§8.1): round down, minimum 1. 116 pub fn night_max_range(weapon: WeaponClass, ae: bool) - > u8 { 117 let day = max_day_range(weapon, ae); 118 if day <= 1 { 1 } else { day / 2 }</pre>
omdurman-rules/src/lib.rs:1511 GH:omdurman-rules/src/lib.rs:1511	effective_movement_at_night	<pre>1509 /// Apply night-turn movement halving for Anglo- Egyptian units (§8.1): all 1510 /// Anglo-Egyptian movement allowances are halved (round down). 1511 pub fn effective_movement_at_night(1512 allowance: MovementAllowance, 1513 player: Player,</pre>

Covered by tests: [night_max_ranges](#) [night_max_ranges_remaining](#) [ae_rifle_at_night_matches_rulebook_example](#) [max_day_range_all_combos](#) [night_movement_overlay_allowance_halved](#)

§8.2 – Dervish Desertion Roll

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:226 GH:omdurman-rules/src/effects.rs:226	DervishDesertion	<pre> 224 /// the effect. The Khalifa, gunboats, artillery, and forts may not be 225 /// chosen. 226 DervishDesertion { 227 roll: DieRoll, 228 deserters: Vec<UnitId>, </pre>
omdurman-rules/src/turn_track.rs:63 GH:omdurman-rules/src/turn_track.rs:63	DervishDesertion	<pre> 61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the first night turn. 63 DervishDesertion, 64 /// Dervish reinforcements are available. 65 DervishReinforcements, </pre>
omdurman-rules/src/turn_track.rs:60 GH:omdurman-rules/src/turn_track.rs:60	TurnEvent	<pre> 58 /// Special events that occur on specific turns (rulebook §8.2, §9.112, §9.113). 59 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 60 pub enum TurnEvent { 61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the first night turn. </pre>

§9 – The Scenarios

§9.1 – The Campaign Game

descriptive

manual page unknown

The Campaign Game

Covered by tests: `campaign_has_no_fixed_placements` `scenario_maps_to_board`

§9.2 – The Historical Scenario

descriptive

manual page unknown

The Historical Scenario

Players should note that the historical scenario is an exercise in futility for the Dervish player. It is, however, an interesting demonstration of the absolute imbecility of the Khalifa's generalship and vividly shows the superiority of entrenched firepower over traditional tribal arms in the colonial period.

Covered by tests: `start_game_scenario_selects_board` `remove_deployed_unit_happy_path`

§9.3 – Bonus Game: Fall of Khartoum

out-of-scope

manual page unknown

Bonus Game: FALL OF KHARTOUM

Covered by tests: `remove_deployed_unit_happy_path`

§9.11 – Set Up (Campaign)

out-of-scope

manual page unknown

Set Up

§9.12 – Scenario Length (Campaign)

implemented

manual page unknown

Scenario Length

6:00 am, Sept. 1 through 8:00 am, Sept. 3, 22 Game Turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:11 GH:omdurman-rules/src/turn_track.rs:11	GameTime	<pre>9 /// starts at one of these twelve times. 10 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 11 pub enum GameTime { 12 SixAM, 13 EightAM,</pre>

omdurman-rules/src/turn_track.rs:47 GH:omdurman-rules/src/turn_track.rs:47	TurnEntry	<pre>45 /// A single entry on the Turn Record Track (rulebook §9.12, §9.22). 46 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 47 pub struct TurnEntry { 48 /// 1-based turn number. 49 pub turn: u8,</pre>
omdurman-rules/src/turn_track.rs:95 GH:omdurman-rules/src/turn_track.rs:95	CAMPAIGN_TURN_TRACK	<pre>93 /// is turn 9, which carries the once-per-game Dervish Desertion Roll (§8.2) -- 94 /// the printed track prints "Dervish Desertion Roll / NIGHT" on that cell. 95 pub const CAMPAIGN_TURN_TRACK: [TurnEntry; 22] = [96 // Row 1, left->right: Sept 1, 6 am -> 8 pm, then the first NIGHT. 97 entry(1, SixAM, Day, TurnEvent::None),</pre>
omdurman-rules/src/turn_track.rs:242 GH:omdurman-rules/src/turn_track.rs:242	TurnLabel	<pre>240 /// printed cell; `Blank` is for unused positions in the 9x3 grid. 241 #[derive(Serialize, Deserialize, Debug, Clone, Copy, PartialEq, Eq)] 242 pub enum TurnLabel { 243 Blank, 244 /// "SEPT. 1\n6:00 am" -- day header plus time (turn 1).</pre>
omdurman-rules/src/turn_track.rs:340 GH:omdurman-rules/src/turn_track.rs:340	turn_marker_pixel	<pre>338 /// 339 /// Rows 0 and 1 use all 9 columns; row 2 uses only columns 0-3. 340 pub fn turn_marker_pixel(track: &omdurman_types::CampaignTurnTrack, turn: u8) -> (f32, f32) { 341 let cell_w = track.w / 9.0; 342 let cell_h = track.h / 3.0;</pre>

Covered by tests: [campaign_track_22_turns](#) [desertion_on_first_night](#) [campaign_track_label_and_day_night_agree](#) [game_time_display_all_variants](#) [turn_label_display](#) [turn_label_out_of_range_is_none](#) [turn_marker_pixel_row_0_left_to_right](#) [turn_marker_pixel_row_1_right_to_left](#) [turn_marker_pixel_rows_are_stacked](#) [game_over_after_campaign_turns](#)

§9.13 – Special Rules (Campaign)

[descriptive](#)

manual page unknown

Special Rules
None.

§9.14 – Victory Conditions (Campaign)

[implemented](#)

manual page unknown

Victory Conditions The Mahdi’s Tomb in Omdurman was not only the tallest structure in the entire Sudan in 1898, it was also a Dervish holy shrine. Its loss or destruction would be a severe blow to the Mahdist cause. It is accordingly assigned 25 victory points which are awarded to the player who controls it at the conclusion of play. The Dervish player controls it at the start of play. As a tactical note, the Anglo-Egyptian player will find a decisive victory almost impossible unless he takes the Mahdi’s Tomb from the Dervish player. To take the Tomb hex, it must be occupied by one British leader plus any one non-“Friendlies” Anglo-Egyptian combat unit (both undisrupted) at the conclusion of play. Additional victory points are awarded as follows: **Dervish Player receives:** • 10 pts: each British leader eliminated

(see manual for full text)

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1157 GH:omdurman-rules/src/lib.rs:1157	VpSource	<pre>1155 /// the manual and the engine. 1156 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1157 pub enum VpSource { 1158 // ----- Anglo-Egyptian player receives: 1159 /// Mahdi's Tomb control at conclusion of play (\$9.14).</pre>
omdurman-rules/src/lib.rs:1183 GH:omdurman-rules/src/lib.rs:1183	VpSource::points	<pre>1181 impl VpSource { 1182 /// VP awarded to `who_scores()` (rulebook §9.14). 1183 pub fn points(self) -> VictoryPoints { 1184 match self { 1185 VpSource::MahdisTomb => VictoryPoints::new(25),</pre>
omdurman-rules/src/lib.rs:1198 GH:omdurman-rules/src/lib.rs:1198	VpSource::who_scores	<pre>1196 1197 /// Which player receives these victory points (rulebook §9.14). 1198 pub fn who_scores(self) -> Player { 1199 match self { 1200 VpSource::MahdisTomb</pre>
omdurman-rules/src/lib.rs:1233 GH:omdurman-rules/src/lib.rs:1233	VictoryLedger	<pre>1231 /// Cumulative victory ledger for one scenario (rulebook §9.14). 1232 #[derive(Serialize, Deserialize, Clone, Debug, Default)] 1233 pub struct VictoryLedger { 1234 pub events: Vec<VpEvent>, 1235 }</pre>
omdurman-rules/src/lib.rs:1239 GH:omdurman-rules/src/lib.rs:1239	VpEvent	<pre>1237 /// A single victory-point scoring event (rulebook §9.14). 1238 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1239 pub struct VpEvent { 1240 pub turn: GameTurnIndex, 1241 pub source: VpSource,</pre>
omdurman-rules/src/lib.rs:1246 GH:omdurman-rules/src/lib.rs:1246	VictoryLedger::total_for	<pre>1244 impl VictoryLedger { 1245 /// Total victory points earned by a given player (rulebook §9.14). 1246 pub fn total_for(&self, player: Player) -> VictoryPoints { 1247 VictoryPoints(1248 self.events</pre>
omdurman-rules/src/lib.rs:1258 GH:omdurman-rules/src/lib.rs:1258	VictoryLedger::superiority	<pre>1256 /// Net superiority: positive = Anglo-Egyptian ahead, negative = Dervish ahead 1257 /// (rulebook §9.14). 1258 pub fn superiority(&self) -> VictoryPoints { 1259 VictoryPoints(self.total_for(Player::AngloEgyptian).value() - self.total_for(Player::Dervish).value()) 1260 }</pre>
omdurman-rules/src/lib.rs:1276 GH:omdurman-rules/src/lib.rs:1276	CampaignVictoryLevel	<pre>1274 /// Campaign-game victory levels (§9.14). 1275 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1276 pub enum CampaignVictoryLevel { 1277 Draw, 1278 Marginal(Player),</pre>
omdurman-rules/src/lib.rs:1285 GH:omdurman-rules/src/lib.rs:1285	CampaignVictoryLevel::from_superiority	<pre>1283 impl CampaignVictoryLevel { 1284 /// Assign a level from the net superiority (\$9.14). 1285 pub fn from_superiority(s: VictoryPoints) -> Self { 1286 let net = s.0;</pre>

		<code>1287 // Positive -> Anglo-Egyptian thresholds: 15/30/50</code>
omdurman-rules/src/effects.rs:5365 GH:omdurman-rules/src/effects.rs:5365	score_elimination	<code>5363 5364 /// Score victory points for eliminating a unit (rulebook §9.14). 5365 pub fn score_elimination(state: &mut GameState, unit_id: UnitId, _owner: Player) { 5366 if let Some(unit) = state.find_unit(unit_id) { 5367 let identity = unit.profile.identity;</code>
omdurman-rules/src/lib.rs:254 GH:omdurman-rules/src/lib.rs:254	VictoryPoints	<code>252 /// (rulebook §9.14). 253 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug, Default)] 254 pub struct VictoryPoints(i32); 255 256 impl VictoryPoints {</code>

Covered by tests: [friendlies_bank_scores_by_side](#) [mahdis_tomb_not_scored_without_a_leader](#)
[mahdis_tomb_scores_for_anglo_egyptian_when_held](#) [vp_source_attributes](#)

§9.21 – Set Up (Historical)

out-of-scope
manual page unknown

Set Up

§9.22 – Scenario Length (Historical)

implemented
manual page unknown

Scenario Length

6:00 am, September 2 through 12:00 noon, September 2. Four game turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:132 GH:omdurman-rules/src/turn_track.rs:132	HISTORICAL_TURN	<code>130 131 /// Historical scenario track (§9.22 -- 4 turns, Sept 2 6:00 am -> 12:00 pm). 132 pub const HISTORICAL_TURN_TRACK: [TurnEntry; 4] = [133 TurnEntry { 134 turn: 1,</code>

Covered by tests: [historical_turn_all_four_turns](#)

§9.23 – Special Rule: The Zariba

implemented
manual page unknown

Special Rule: “The Zariba”

****9.231)** Thorn hedge hexsides:****** –2 to die roll on all Dervish fire attacks; may not melee across in either direction; may not advance after combat across in either direction.

****9.232)** Trench hexsides:****** –4 to die roll on all Dervish fire attacks vs. entrenched units only; –2 (instead of +2) melee modifier to Dervish units melee attacking an entrenched unit; entrenched units may be fired “over” in both directions (i.e. they do not block line of sight); units are considered “entrenched” if they are directly adjacent to (and on the Nile River side of) a trench hexside.

(see manual for full text)

See also: §9.231, §9.232

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:925 GH:omdurman-rules/src/lib.rs:925	ZaribaThornHedge	<pre>923 Terrain(i16), 924 /// -2 thorn-hedge defensive modifier (§9.231). 925 ZaribaThornHedge, 926 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 927 /// units (those Nile-side of the trench hexside).</pre>
omdurman-rules/src/lib.rs:928 GH:omdurman-rules/src/lib.rs:928	ZaribaTrenchEntrenched	<pre>926 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 927 /// units (those Nile-side of the trench hexside). 928 ZaribaTrenchEntrenched, 929 } 930 </pre>
omdurman-types/src/lib.rs:195 GH:omdurman-types/src/lib.rs:195	ZaribaThornHedge	<pre>193 Crest, 194 /// Historical-scenario thorn-hedge segment of the Zariba (§9.231). 195 ZaribaThornHedge, 196 /// Historical-scenario trench segment of the Zariba (§9.232). 197 ZaribaTrench,</pre>
omdurman-types/src/lib.rs:197 GH:omdurman-types/src/lib.rs:197	ZaribaTrench	<pre>195 ZaribaThornHedge, 196 /// Historical-scenario trench segment of the Zariba (§9.232). 197 ZaribaTrench, 198 /// One of the two end hexsides of a Zariba trench segment that connect to 199 /// the Nile River (§9.233). Units may only enter/ leave the Zariba via</pre>

§9.24 – Victory Conditions (Historical)

implemented

manual page unknown

Victory Conditions

Victory Levels are based solely on eliminating enemy units while conserving your own force as much as possible.

| Victory Level | Anglo-Egyptian Player (Dervish units eliminated) | Dervish Player (Anglo-Egyptian units eliminated)
| |—|—|—| | 5 — DECISIVE | 100+ | 30+ | | 4 — STRATEGIC | 60–99 | 15–29 | | 3 — TACTICAL | 45–59 | 10–14 | | 2 — MARGINAL | 30–44 | 5–9 | | 1 — DRAW | 0–29 | 0–4 |

The lower value victory level is then subtracted from the higher level to determine a player's net victory. For example, if the Anglo-Egyptian player eliminates 104 Dervish units (decisive victory) but loses 18 units doing it (Dervish Strategic), the Anglo-Egyptian player only nets out with a draw (decisive worth 5 minus strategic worth 4 = 1, draw).

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1315 GH:omdurman-rules/src/lib.rs:1315	HistoricalVictoryLevel	<pre>1313 /// draw"). 1314 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug)] 1315 pub enum HistoricalVictoryLevel { 1316 Draw = 1, 1317 Marginal = 2,</pre>

Covered by tests: `historical_victory_level_for_dervish` `historical_victory_level_for_anglo_egyptian`

§9.31 – Bonus game map

`out-of-scope`
manual page unknown

Only the small FALL OF KHARTOUM scenario map is used for this game.

§9.32 – Set Up (Bonus)

`out-of-scope`
manual page unknown

Set Up

§9.33 – Scenario Length (Bonus)

`implemented`
manual page unknown

Scenario Length
Variable, see victory conditions (9.35). Rarely lasts five turns.

See also: §9.35

File	Symbol	Code Snippet
<code>omdurman-rules/src/turn_track.rs:172</code> GH:omdurman-rules/src/turn_track.rs:172	<code>FALL_OF_KHARTOUM</code>	<pre>170 /// (the rulebook fixes none); only `day_night` is rule-bearing (night halves 171 /// Anglo-Egyptian movement and ranges and bars howitzer fire, §8.1). 172 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [173 TurnEntry { 174 turn: 1,</pre>

Covered by tests: `fall_of_khartoum_turn_one_is_night` `fall_of_khartoum_turns_3_to_8_are_day`

§9.34 – Special Rules (Bonus)

`out-of-scope`
manual page unknown

Special Rules

§9.35 – Victory Conditions (Bonus)

`implemented`
manual page unknown

Victory Conditions
Victory is determined by how many turns it takes the Dervish player to eliminate the GORDON leader unit and how many Dervish units are eliminated:

- Dervish decisive: eliminate GORDON turn four or sooner.
- Dervish tactical: eliminate GORDON turn five.
- Dervish marginal: eliminate GORDON turn six.

- British marginal: GORDON survives end of turn six.
- British tactical: GORDON survives end of turn seven.
- British decisive: GORDON survives end of turn eight.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1356 GH:omdurman-rules/src/lib.rs:1356	FoKVictoryLevel	<pre> 1354 /// negative) so the loss penalty is a simple shift toward the British end. 1355 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug)] 1356 pub enum FoKVictoryLevel { 1357 DervishDecisive = -3, 1358 DervishTactical = -2,</pre>

Covered by tests: [fok_victory_level_worked_example](#) [fok_victory_level_gordon_died_early](#) [fok_victory_level_late_gordon_death](#) [fok_victory_level_gordon_survived](#)

§9.111 – Dervish set up (Campaign)

implemented

manual page unknown

Dervish player sets up first, moves second.

- Isa Zachneih infantry unit: anywhere on the east bank, in or south of El Debeba.
- KHALIFA ABDULLAH: in the walled city of Omdurman, in either palace hex.
- 3 artillery units, and all Taiasha units: anywhere in the walled city of Omdurman.
- 17 forts: anywhere on the mapsheet south of the Khor Shambat on the west bank, and/or south of all Halfaya hut hexes on the east bank and Nile River islands.
- 2 gunboats: any south edge Nile River hexes.

****9.112) Dervish reinforcements:**** all reinforcements enter on the west edge of the mapsheet, south of the Khor Shambat. Each unit pays the terrain cost of the hex through which it enters, no matter how many units enter through

(see manual for full text)

See also: §9.112

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1044 GH:omdurman-rules/src/effects.rs:1044	setup_complete	<pre> 1042 /// currently shares the same "both sides deployed" gate; when a scenario 1043 /// needs a different minimum, branch on `self.scenario` here. 1044 pub fn setup_complete(&self) -> Result<(), RuleError> { 1045 let has = player { 1046 self.units</pre>

Covered by tests: [hadendowa_first_cell_is_isa_zachneih](#) [campaign_deployment_is_boat_land_exclusive](#) [campaign_setup_rejects_non_initial_force](#)

§9.112 – Dervish reinforcements (Campaign)

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:218 GH:omdurman-	PlaceReinforcements	<pre> 216 // -- Reinforcement / placement ----- 217 /// Place reinforcements onto the map (rulebook</pre>

rules/src/ effects.rs:218		<pre> \$9.112, \$9.113). 218 PlaceReinforcements(Vec<UnitPlacement>), 219 220 // -- Scenario-specific ----- </pre>
omdurman- rules/src/ effects.rs:4710 GH:omdurman- rules/src/ effects.rs:4710	apply_place_reinforce	<pre> 4708 4709 /// Place reinforcements onto the map (rulebook \$9.112, \$9.113). 4710 pub fn apply_place_reinforcements(4711 state: &mut GameState, 4712 placements: &[UnitPlacement], </pre>
omdurman- rules/src/reinforcements.rs:74 GH:omdurman- rules/src/reinforcements.rs:74	dervish_campaign_sched	<pre> 72 /// All reinforcements enter on the west edge, south of the Khor Shambat. 73 /// Each unit pays terrain cost of the hex it enters through. 74 pub fn dervish_campaign_schedule() -> ReinforcementSchedule { 75 ReinforcementSchedule { 76 player: Player::Dervish, </pre>
omdurman- types/src/ lib.rs:509 GH:omdurman- types/src/lib.rs:509	Location	<pre> 507 /// Named map landmarks (rulebook mapsheet, \$9.111, \$9.113, \$9.212 scenarios). 508 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 509 pub enum Location { 510 FortMakran, 511 NorthFort, </pre>
omdurman- types/src/ lib.rs:589 GH:omdurman- types/src/lib.rs:589	SetupLetter	<pre> 587 /// Each letter marks a specific hex where a Dervish leader is placed. 588 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 589 pub enum SetupLetter { 590 Y, 591 K, </pre>
omdurman- types/src/ lib.rs:717 GH:omdurman- types/src/lib.rs:717	Faction	<pre> 715 /// = the printed designation). For Friendlies, the editor sets 716 /// `Some(BrigadeId::friendlies())`. 717 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 718 pub enum Faction { 719 Dervish { tribe: DervishTribe }, </pre>

Covered by tests: **dervish_schedule_has_three_waves** **dervish_wave_one_has_baggaara_and_three_leaders**
dervish_wave_two_has_hadendowa **dervish_wave_three_is_all_remaining** **wave_for_turn_returns_correct_wave**
campaign_reinforcements_gate_by_wave

§9.113 – Anglo-Egyptian reinforcements (Campaign)

implemented

manual page unknown

The Anglo-Egyptian player moves first. There are no Anglo-Egyptian units on the mapsheet at start. The GORDON unit is not used in this scenario.

- The leader units KITCHENER, GATACRE, and HUNTER may enter anytime during the first four game turns and do not count against the 12 units per turn limit. All three leaders must be in play by the end of turn four!
- All gunboats enter through any north edge Nile River hex, paying one movement point for the first hex entered. The “Friendlies” brigade enters through the Abu Alim hut hex on the east bank, paying eight movement points per unit. All other Anglo-Egyptian units enter through the west bank “ANGLO-EGYPTIAN ENTRANCE AREA”, each unit paying one movement point to enter the mapsheet.
- Turn 1) Any three gunboats; “Friendlies” brigade; Egyptian Cavalry; Horse Artillery; and two infantry brigades

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/reinforcements.rs:131 GH:omdurman-rules/src/reinforcements.rs:131	anglo_egyptian_campaign	<pre> 129 /// - "Friendlies" enter via Abu Alim hut on the east bank (8 MP per unit). 130 /// - All other AE units enter via the Anglo-Egyptian Entrance Area (1 MP). 131 pub fn anglo_egyptian_campaign_schedule() -> ReinforcementSchedule { 132 let free_leaders = vec![133 CampaignLeader::British(BritishLeader::Kitchener), </pre>

Covered by tests: `anglo_egyptian_schedule_has_four_waves` `anglo_egyptian_leaders_available_each_wave` `anglo_egyptian_turn_four_is_all_remaining` `campaign_reinforcement_cap_and_double_entry` `campaign_gunboats_quota_three_per_turn` `reinforcement_rejected_onto_enemy_occupied_hex`

§9.211 – Anglo-Egyptian set up first, moves second (Historical)

implemented

manual page unknown

The Anglo-Egyptian player sets up first, and moves second:

- Not in play: GORDON leader unit, “Friendlies” brigade.
- Gunboats start in any Nile River hexes adjacent to the Zariba.
- Camel Corps, Egyptian Cavalry, and Horse Artillery start in the village of Kerreri hut hexes.
- All remaining Anglo-Egyptian units set up in the 13 hexes of the Zariba.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:2911 GH:omdurman-rules/src/effects.rs:2911	first_player	<pre> 2909 2910 /// The player who moves first in a scenario (§4, §9.113, §9.212, §9.322). 2911 pub fn first_player(scenario: Scenario) -> Player { 2912 match scenario { 2913 Scenario::Campaign => Player::AngloEgyptian, </pre>

Covered by tests: `historical_setup_rejects_not_in_play_units`

§9.212 – Dervish set up (Historical) – deployment zones and leader-to-hex mapping

implemented

manual page unknown

The Dervish player sets up second, and moves first.

- Not in play: Isa Zachneih, gunboats, and forts.
- All Dervish units must be set up out of the line of sight of all Anglo-Egyptian units.
- Dervish leaders start on the lettered hexes:
 - A: Ali Wad Helu
 - D: Sheik El Din
 - Y: Yakub
 - K: Khalifa Abdullah
 - S: Sherif
 - O: Osman Digna

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/	in_deployment_zone	<pre> 1152 /// plan / UI rather than this hex predicate. Documented, not silently </pre>

effects.rs:1154 GH:omdurman-rules/src/effects.rs:1154		<pre> 1153 /// dropped. 1154 pub fn in_deployment_zone(&self, player: Player, hex: HexCoord, is_boat: bool) -> bool { 1155 // No board attached -> permissive (unit tests, unbound session). 1156 if self.board.terrain.is_empty() { </pre>
omdurman-rules/src/lib.rs:410 GH:omdurman-rules/src/lib.rs:410	DervishLeader::setup_letter	<pre> 408 /// (§9.212): A→Ali Wad Helu, D→Sheik El Din, Y→Yakub, K→Khalifa Abdullah, 409 /// S→Sherif, O→Osman Digna. Inverse of [`dervish_leader_for_setup_letter`]. 410 pub fn setup_letter(self) -> SetupLetter { 411 match self { 412 DervishLeader::AliWadHelu => SetupLetter::A, </pre>
omdurman-rules/src/lib.rs:426 GH:omdurman-rules/src/lib.rs:426	dervish_leader_for_setup_letter	<pre> 424 /// inherent impl here, so the mapping is a free function -- the bijective 425 /// inverse of [`DervishLeader::setup_letter`]. 426 pub fn dervish_leader_for_setup_letter(letter: SetupLetter) -> DervishLeader { 427 match letter { 428 SetupLetter::A => DervishLeader::AliWadHelu, </pre>

Covered by tests: `deploy_rejected_outside_zone` `embedded_leaders_resolve_from_their_host_section` `historical_places_all_six_leaders_when_anchors_present` `missing_anchor_is_reported_not_dropped_silently` `setup_letter_dervish_leader_roundtrip` `setup_letter_to_dervish_leader_known_values` `historical_setup_rejects_not_in_play_units`

§9.231 – Thorn hedge hexsides

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:925 GH:omdurman-rules/src/lib.rs:925	ZaribaThornHedge	<pre> 923 Terrain(i16), 924 /// -2 thorn-hedge defensive modifier (§9.231). 925 ZaribaThornHedge, 926 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 927 /// units (those Nile-side of the trench hexside). </pre>
omdurman-types/src/lib.rs:195 GH:omdurman-types/src/lib.rs:195	ZaribaThornHedge	<pre> 193 Crest, 194 /// Historical-scenario thorn-hedge segment of the Zariba (§9.231). 195 ZaribaThornHedge, 196 /// Historical-scenario trench segment of the Zariba (§9.232). 197 ZaribaTrench, </pre>

Covered by tests: `zariba_fire_penalties_apply_to_dervish_fire_only`

§9.232 – Trench hexsides

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:928 GH:omdurman-rules/src/lib.rs:928	ZaribaTrenchEntrenched	<pre> 926 /// -4 trench defensive modifier (§9.232). Only applies vs. "entrenched" 927 /// units (those Nile-side of the trench hexside). 928 ZaribaTrenchEntrenched, 929 } 930 </pre>

omdurman-types/src/lib.rs:197 GH:omdurman-types/src/lib.rs:197	ZaribaTrench	<pre> 195 ZaribaThornHedge, 196 /// Historical-scenario trench segment of the Zariba (§9.232). 197 ZaribaTrench, 198 /// One of the two end hexsides of a Zariba trench segment that connect to 199 /// the Nile River (§9.233). Units may only enter/ leave the Zariba via </pre>
---	---------------------	--

§9.233 – Zariba entry/exit costs

implemented

manual page unknown

Units may only enter and/or leave the Zariba via the two end hexsides that connect to the Nile River, paying +2 movement points to cross (Exception: advance after combat across an entrenched hexside).

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:247 GH:omdurman-types/src/lib.rs:247	blocks_movement	<pre> 245 /// `omdurman-rules`). The trench *end* variants are therefore intentionally 246 /// not blocking. 247 pub fn blocks_movement(self) -> bool { 248 matches!(249 self, </pre>
omdurman-rules/src/board.rs:280 GH:omdurman-rules/src/board.rs:280	zariba_entry_surcharge	<pre> 278 /// movement points to cross"). Returns 2 when the edge between `from` and 279 /// `to` is one of the two trench ends, else 0. 280 pub fn zariba_entry_surcharge(&self, from: HexCoord, to: HexCoord) -> i16 { 281 match self.hexside_between(from, to) { 282 Some(k) if k.is_zariba_trench_end() => 2, </pre>
omdurman-rules/src/effects.rs:1639 GH:omdurman-rules/src/effects.rs:1639	movement_cost_for	<pre> 1637 /// 1638 /// §5.42: entering or leaving an enemy ZOC adds no MP cost. 1639 fn movement_cost_for(&self, unit: &UnitPlacement, path: &[HexCoord]) -> Option<MovementPoints> { 1640 if path.is_empty() self.board.terrain.is_empty() { 1641 return None; </pre>
omdurman-rules/src/effects.rs:3122 GH:omdurman-rules/src/effects.rs:3122	apply_move_unit	<pre> 3120 /// the true terrain cost (§5.11) and enforces gunboat upstream/downstream 3121 /// allowances (§5.24); otherwise it falls back to the caller-supplied `cost`. 3122 pub fn apply_move_unit(3123 state: &mut GameState, 3124 unit_id: UnitId, </pre>

Covered by tests: [zariba_end_hexside_costs_extra_mp](#) [zariba_thorn_hedge_blocks_movement](#)

§9.321 – British set up (Bonus)

implemented

manual page unknown

The British player sets up first, moves second:

- General GORDON leader unit in the palace.
- Two old style (unnamed) gunboats in any Nile River hexes.
- Set up in any building or hut hexes of Khartoum, Forts Makran and/or Buri, and/or adjacent to any wall hex:
 - one Egyptian Battalion artillery unit
 - two British infantry units (represents Caucasian troops)

- ▶ three Egyptian infantry units (represents Cairo “Bazouks”)
- ▶ four Sudan infantry units (represents Sudanese blacks)
- ▶ four “Friendlies” units (represents the Shaggyeh)

(see manual for full text)

File	Symbol	Code Snippet
omdurman-app/src/scenario_setup.rs:115 GH:omdurman-app/src/scenario_setup.rs:115	FALL_OF_KHARTOUM	<pre> 113 /// GORDON is the "GEN. GORDON" counter at BritishBoats (3,1); the North Fort 114 /// uses a campaign HadendowaForts counter (one of the spare fort sprites). 115 const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[116 FixedPlacement { 117 section: SectionName::BritishBoats,</pre>

Covered by tests: [confirm_ready_rejected_below_scenario_target](#) [fall_of_khartoum_places_gordon_in_the_palace](#) [fall_of_khartoum_reports_missing_palace](#) [fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes](#) [fok_ae_gunboat_deploys_only_on_nile](#) [fok_ae_land_unit_rejected_on_nile](#) [fok_setup_complete_requires_full_oob](#) [deploy_via_real_sprite_resolution_matches_engine](#) [british_boats_named_vs_old_gunboat_detection](#) [fok_order_of_battle_british](#) [fok_caps_bind_across_counter_variants](#)

§9.322 – Dervish enters turn one (Bonus)

implemented

manual page unknown

Dervish player moves first: enters turn one through any hexes on the south or east edge of the map.

- 32 Mulazmin units (represents combined forces of Wad El Nejumi, Abu Girgeh, and Sheik El Obeid)
- 2 Hadendowa; 6 Kehena; 5 Degheim (represents Mahdi’s combined west bank forces)
- 3 Dervish artillery units.

File	Symbol	Code Snippet
omdurman-rules/src/unit_profiles.rs:317 GH:omdurman-rules/src/unit_profiles.rs:317	ali_wad_helu	<pre> 315 /// (3-6-9) -- the Degheim force of §9.322, printed on Baggara-backed 316 /// sprites. 317 fn ali_wad_helu(col: u32, row: u32) -> Option<Classification> { 318 match (col, row) { 319 (0, 0) => dervish_leader(DervishLeader::AliWadHelu),</pre>
omdurman-types/src/lib.rs:807 GH:omdurman-types/src/lib.rs:807	sections_for_picker	<pre> 805 /// (row 1) + 5 Degheim (row 0, cols 1-5) tribes. KhalifaAbdullah 806 /// provides the 3 artillery, and HadendowaForts supplies the 807 /// Dervish-controlled North Fort sprite (§9.344). 808 pub fn sections_for_picker(self) -> Option<&'static [SectionName]> { 809 match self {</pre>

Covered by tests: [ali_wad_helu_block_resolves_leader_and_degelim_tribes](#) [fok_setup_complete_requires_full_oob](#) [fok_dervish_land_unit_rejected_on_nile](#) [fok_picker_allowlist_has_dervish_entry_force_blocks](#) [fok_order_of_battle_dervish](#) [fok_dervish_east_edge_on_diamond_board](#)

§9.341 – Turn 1 is always a night turn (Bonus)

implemented

manual page unknown

Turn 1 is always a night turn (see 8.1).

See also: §8.1

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:172 GH:omdurman-rules/src/turn_track.rs:172	FALL_OF_KHARTOUM	<pre>170 /// (the rulebook fixes none); only `day_night` is 171 rule-bearing (night halves 172 howitzer fire, §8.1). 173 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = 174 [175 TurnEntry { 176 turn: 1,</pre>

Covered by tests: [fall_of_khartoum_turn_one_is_night](#)

§9.342 – All hexes are playable, including half hexes (Bonus)

implemented

manual page unknown

All hexes are playable, including hexes showing up half or less.

File	Symbol	Code Snippet
omdurman-rules/src/board_data.rs:33 GH:omdurman-rules/src/board_data.rs:33	fall_of_khartoum_map	<pre>31 /// The Campaign board (also used by the Historical 32 scenario, §9.1/§9.2). 33 pub fn campaign_map_data() -> MapData { 34 CAMPAIGN.get_or_init(35 parse(CAMPAIGN_ROM).clone() 36) 37 }</pre>

§9.343 – Both players use the Dervish Range Effects Table (Bonus)

implemented

manual page unknown

Both players must use the Dervish Range Effects Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:3359 GH:omdurman-rules/src/effects.rs:3359	range_band_for	<pre>3357 /// in FALL OF KHARTOUM *both* players use the Dervish 3358 Range Effects Table 3359 /// (§9.343). 3360 pub fn range_band_for(3361 scenario: Scenario, 3362 player: Player,</pre>

§9.344 – Dervish controls the North Fort (Bonus)

implemented

manual page unknown

The Dervish player controls the “North Fort” and may fire its guns.

File	Symbol	Code Snippet
------	--------	--------------

omdurman-app/ src/scenario_setup.rs:115 GH:omdurman-app/src/scenario_setup.rs:115	FALL_OF_KHARTOUM	<pre> 113 /// GORDON is the "GEN. GORDON" counter at British_Boats (3,1); the North Fort 114 /// uses a campaign HadendowaForts counter (one of the spare fort sprites). 115 const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[116 FixedPlacement { 117 section: SectionName::BritishBoats,</pre>
omdurman-rules/src/ effects.rs:2189 GH:omdurman-rules/src/ effects.rs:2189	hex_has_enemy_fort	<pre> 2187 /// may neither occupy an enemy fort nor advance after combat into one 2188 /// (forts are never captured -- only destroyed, \$6.62/\$6.53/\$7.6). 2189 pub fn hex_has_enemy_fort(&self, hex: HexCoord, mover: Player) -> bool { 2190 self.units.iter().any(u { 2191 u.position == hex</pre>

Covered by tests: `fall_of_khartoum_places_gordon_in_the_palace` `fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes` `placement_done_gate_matches_by_identity_not_allocated_id` `fok_order_of_battle_dervish` `fok_picker_allowlist_has_dervish_entry_force_blocks`

§9.345 – Gunboat White Nile <-> Blue Nile crossing (Bonus)

implemented

manual page unknown

The British gunboats may move from the White Nile to the Blue Nile and vice-versa at an off-board movement cost of six “upstream” movement points.

File	Symbol	Code Snippet
omdurman-rules/src/ effects.rs:2173 GH:omdurman-rules/src/ effects.rs:2173	is_nile_mouth_crossing	<pre> 2171 /// must be named on the board, else this is `false` and the move falls 2172 /// through to the ordinary contiguous-Nile rules. 2173 pub fn is_nile_mouth_crossing(&self, from: HexCoord, to: HexCoord) -> bool { 2174 let white = self 2175 .board</pre>
omdurman-types/src/ lib.rs:525 GH:omdurman-types/src/lib.rs:525	Location::WhiteNileMouth	<pre> 523 /// The off-board mouth of the White Nile branch (FALL OF KHARTOUM §9.345) -- 524 /// a British gunboat may cross to the Blue Nile mouth for 6 upstream MP. 525 WhiteNileMouth, 526 /// The off-board mouth of the Blue Nile branch (FALL OF KHARTOUM §9.345). 527 BlueNileMouth,</pre>

§9.346 – GORDON immobile, eliminated only at the Palace (Bonus)

implemented

manual page unknown

The GORDON leader unit starts in the palace and may not move during the scenario. He may only be eliminated by a Dervish unit passing through or occupying the palace hex (as normal movement or as advance after combat).

File	Symbol	Code Snippet
omdurman-rules/src/ effects.rs:3040 GH:omdurman-rules/src/ effects.rs:3040	check_gordon_palace	<pre> 3038 /// after combat). Records the turn (which fixes the §9.35 victory level) and 3039 /// ends the game. A no-op outside FoK, or once GORDON is already gone. 3040 pub fn check_gordon_palace(state: &mut GameState) { 3041 if state.scenario != Scenario::FallOfKhartoum </pre>

		<pre>state.gordon_eliminated_turn.is_some() { 3042 return;</pre>
omdurman-rules/src/lib.rs:646 GH:omdurman-rules/src/lib.rs:646	UnitIdentity::is_gordon	<pre>644 /// Whether this is the GORDON leader unit (§9.32, §9.346) -- the immobile 645 /// palace defender whose elimination ends FALL OF KHARTOUM (§9.35). 646 pub fn is_gordon(&self) -> bool { 647 matches!(648 self,</pre>

Covered by tests: [gordon_is_an_immobile_british_leader](#) [gordon_survives_means_no_elimination](#) [fall_of_khartoum_places_gordon_in_the_palace](#)

10/10 implemented

§10 – Optional Rules

§10 – Optional Rules

implemented

manual page unknown

Optional Rules (Campaign game only)

It is suggested that the most intriguing employment of the following two options is to permit the Dervish player to have either one or the other, but the Anglo-Egyptian player doesn't know which one until he stumbles onto it. Players are advised that the employment of both optionals in the same game is not recommended.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:339 GH:omdurman-rules/src/lib.rs:339	OptionalRule	<pre>337 /// two should be in play (rulebook §10). 338 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 339 pub enum OptionalRule { 340 RiverMines, 341 RiverChain,</pre>

§10.1 – River Mines

implemented

manual page unknown

River Mines

The Khalifa twice tried (unsuccessfully) to submerge a powerful mine in the Nile to sink or damage British gunboats. This option assumes that both attempts were successful.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:236 GH:omdurman-rules/src/effects.rs:236	RiverMine	<pre>234 // -- Optional rules ----- 235 /// River mine resolution (rulebook §10.12). 236 RiverMine { 237 gunboat_id: UnitId, 238 hex: HexCoord,</pre>
omdurman-rules/src/effects.rs:4988 GH:omdurman-rules/src/effects.rs:4988	apply_river_mine	<pre>4986 4987 /// Apply a river-mine resolution (rulebook §10.12). 4988 pub fn apply_river_mine(4989 state: &mut GameState, 4990 gunboat_id: UnitId,</pre>
omdurman-rules/src/lib.rs:1109 GH:omdurman-rules/src/lib.rs:1109	MineResult	<pre>1107 /// British gunboat enters a mined hex. 1108 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1109 pub enum MineResult { 1110 /// Roll 1-4: no effect. 1111 NoEffect,</pre>

§10.2 – River Chain

implemented

manual page unknown

River Chain

The Khalifa also tried (also unsuccessfully) to string a heavy chain across the Nile to stop or slow down the British gunboats. This option assumes the chain was emplaced.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:5084 GH:omdurman-rules/src/effects.rs:5084	apply_place_chain	<pre> 5082 /// Lay (or replace) the river chain during setup (\$10.21). Validated by 5083 /// [<code>GameState::can_place_chain`</code>]. 5084 pub fn apply_place_chain(state: &mut GameState, hexes: &[HexCoord]) -> Result<(), RuleError> { 5085 state.can_place_chain(hexes)?; 5086 state.chain = Some(ChainPlacement { </pre>
omdurman-rules/src/effects.rs:3946 GH:omdurman-rules/src/effects.rs:3946	MAX_CHAIN_HEXES	<pre> 3944 3945 /// Maximum contiguous Nile hexes the river chain may span (\$10.21). 3946 pub const MAX_CHAIN_HEXES: usize = 4; 3947 3948 // </pre>

§10.11 – Secretly record mine hexes

implemented

manual page unknown

Prior to the commencement of play the Dervish player secretly records two Nile River hexes to be mined (the mines may not both be placed in the same hex). These hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:5073 GH:omdurman-rules/src/effects.rs:5073	apply_place_mine	<pre> 5071 /// Lay a river mine during setup (§10.11). Validated by 5072 /// [<code>GameState::can_place_mine`</code>]. 5073 pub fn apply_place_mine(state: &mut GameState, hex: HexCoord) -> Result<(), RuleError> { 5074 state.can_place_mine(hex)?; 5075 state.mines.push(MinePlacement { </pre>
omdurman-rules/src/effects.rs:7404 GH:omdurman-rules/src/effects.rs:7404	GameState::mines	<pre> 7402 state.optional_rules.push(OptionalRule::RiverMines); 7403 7404 state.optional_rules.push(OptionalRule::RiverChain); // Two mines OK, a third rejected. 7405 apply_effect(7406 &mut state, </pre>

Covered by tests: [mine_and_chain_limits_enforced_in_setup](#) [mines_and_chain_require_their_optional_rule](#)

§10.12 – Mine resolution

implemented

manual page unknown

When a British gunboat enters a mined hex, the Dervish player must order it to stop as it has struck a mine. The Dervish player then resolves the effect of the mine's blast by rolling the ten-sided die:

- 1–4: No effect
- 5–7: Gunboat damaged, lost use of its engines and must drift two hexes per turn (with the current) for the rest of the game. No effect on guns or Maxims unless they drift out of range.
- 8–10: Gunboat sunk!

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:236 GH:omdurman-rules/src/effects.rs:236	RiverMine	<pre> 234 // -- Optional rules ----- 235 /// River mine resolution (rulebook §10.12). 236 RiverMine { 237 gunboat_id: UnitId, 238 hex: HexCoord, </pre>
omdurman-rules/src/effects.rs:4988 GH:omdurman-rules/src/effects.rs:4988	apply_river_mine	<pre> 4986 4987 /// Apply a river-mine resolution (rulebook §10.12). 4988 pub fn apply_river_mine(4989 state: &mut GameState, 4990 gunboat_id: UnitId, </pre>

§10.13 – Mines consumed after both rolled for

implemented

manual page unknown

After both mines have been rolled for, no more are available.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:4988 GH:omdurman-rules/src/effects.rs:4988	apply_river_mine	<pre> 4986 4987 /// Apply a river-mine resolution (rulebook §10.12). 4988 pub fn apply_river_mine(4989 state: &mut GameState, 4990 gunboat_id: UnitId, </pre>

§10.14 – Dervish gunboats pass safely

implemented

manual page unknown

The Dervish player's gunboats may pass through the mined hexes with no ill effect (he knows where the mines are).

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:4988 GH:omdurman-rules/src/effects.rs:4988	apply_river_mine	<pre> 4986 4987 /// Apply a river-mine resolution (rulebook §10.12). 4988 pub fn apply_river_mine(4989 state: &mut GameState, 4990 gunboat_id: UnitId, </pre>

§10.21 – Secretly record chain hexes

implemented

manual page unknown

Prior to the commencement of play the Dervish player secretly records a line of river hexes (not exceeding four hexes long) across which the chain is strung. The hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1412	can_place_chain	<pre> 1410 /// Read-only check of a river-chain placement in setup (§10.21): Setup phase 1411 /// and at most [<code>MAX_CHAIN_HEXES</code>] hexes. </pre>

GH:omdurman-rules/src/effects.rs:1412		<pre> 1412 pub fn can_place_chain(&self, hexes: &[HexCoord]) -> Result<(), RuleError> { 1413 self.require_setup_phase()?; 1414 // Optional-rule gate: the chain exists only when the River Chain option </pre>
omdurman-rules/src/effects.rs:5084 GH:omdurman-rules/src/effects.rs:5084	apply_place_chain	<pre> 5082 /// Lay (or replace) the river chain during setup (\$10.21). Validated by 5083 /// [<code>GameState::can_place_chain`</code>]. 5084 pub fn apply_place_chain(state: &mut GameState, hexes: &[HexCoord]) -> Result<(), RuleError> { 5085 state.can_place_chain(hexes)?; 5086 state.chain = Some(ChainPlacement { </pre>

Covered by tests: `mine_and_chain_limits_enforced_in_setup` `mines_and_chain_require_their_optional_rule`

§10.22 – Gunboat stops on chained hex

implemented

manual page unknown

When a British gunboat enters a “chained” river hex it must stop and may move no further that turn.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:1672 GH:omdurman-rules/src/effects.rs:1672	can_move_gunboat	<pre> 1670 /// upstream movement allowance is their maximum for that turn." Chained Nile 1671 /// hexes stop the gunboat (§10.22). 1672 pub fn can_move_gunboat(1673 &self, 1674 unit_id: UnitId, </pre>

§10.23 – Sinking the chain

implemented

manual page unknown

No gunboats (British or Dervish) may cross the chain until it has been sunk by the British player. He may sink the chain by a) having an infantry or cavalry unit spend one complete turn on either riverbank adjacent to a “chained” river hex, or b) firing at the chain with artillery and achieving a 3 or more on the Combat Results Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects.rs:5032 GH:omdurman-rules/src/effects.rs:5032	apply_sink_chain	<pre> 5030 /// Sink the river chain (rulebook §10.23). Marks the placed chain cleared so it 5031 /// no longer stops gunboats (§10.22). 5032 pub fn apply_sink_chain(state: &mut GameState) -> Result<(), RuleError> { 5033 match state.chain.as_mut() { 5034 Some(chain) if !chain.sunk => { </pre>

0/1 implemented

§11 – Historical Notes

§11 – Historical Notes

descriptive

manual page unknown

Historical Notes

In 1881 Mohammed Ahmed Ibn Al-Sayid Abdullah, the son of an obscure carpenter in the hinterlands of the Sudan, proclaimed himself the “Mahdi” — the Messiah of the Islamic faith. His timing was propitious indeed. Since the early 1820’s a corrupt Egypt, with the Sultan of Turkey’s blessing, had incessantly raped the Sudan, taking ivory and flooding the slave markets with some half million captured Sudanese blacks. By 1880, nearly 40,000 Egyptian troops occupied outposts scattered throughout the Sudan, enforcing Egypt’s hold on this lucrative ivory and slave trade and squeezing the native population dry through vicious and corrupt tax officials. All was controlled from Khartoum via the office of Governor General of the Sudan. The title had been held by a succession of individuals, including General Charles Gordon, whose appointment was an attempt to reinstate some rudimentary justice in the Sudan after France and Britain assumed joint political control of a bankrupt Egypt.

By 1881, however, Gordon’s term had expired and a new Governor General, again corrupt and incompetent,
(see manual for full text)

See also: §9.1, §9.2, §9.3

0/1 implemented

Credits

§Credits – Credits

descriptive

manual page unknown

 1/1 implemented

Combat Results Table (shared reference)

§CRT – Combat Results Table (shared by §6.22 fire and §7.7 melee)

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/combat_results_table.rs:8 GH:omdurman-rules/src/combat_results_table.rs:8	FireFactorRow	<pre>6 /// to index into the result matrix. 7 #[derive(serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 8 pub enum FireFactorRow { 9 /// 1-5 factors 10 Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:31 GH:omdurman-rules/src/combat_results_table.rs:31	from_total	<pre>29 impl FireFactorRow { 30 /// Determine which row a given total fire factor falls into (rulebook §6.22). 31 pub fn from_total(total: u16) -> Self { 32 match total { 33 0..=5 => FireFactorRow::Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:69 GH:omdurman-rules/src/combat_results_table.rs:69	combat_results_table	<pre>67 /// D = `Disrupt` (1/2 of target units, round up) 68 /// 1...5 = `Eliminate(n)` (that many units removed) 69 pub fn combat_results_table(row: FireFactorRow, roll: DieRoll) -> CombatResult { 70 use CombatResult::*; 71 use DieRoll::*;</pre>
omdurman-rules/src/lib.rs:989 GH:omdurman-rules/src/lib.rs:989	CombatResult	<pre>987 /// * `--` -- no effect 988 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 989 pub enum CombatResult { 990 NoEffect, 991 Disrupt,</pre>

Covered by tests: [ae_combat_results_table_lowest_is_no_effect](#) [ae_combat_results_table_highest_is_eliminate_5](#) [ae_combat_results_table_progresses_with_roll](#) [ae_combat_results_table_progresses_with_factor](#) [fire_factor_row_boundaries](#) [fire_factor_row_remaining_boundaries](#) [fire_factor_row_index_sequential](#) [crt_all_rows_monotone_non_decreasing](#) [crt_cross_row_monotone_for_each_roll](#) [crt_lowest_row_is_worst_highest_row_is_best](#) [crt_eliminate_never_exceeds_5](#) [crt_every_cell_matches_the_table](#)

0/1 implemented

| Reference – Charts and Tables

§Reference – Charts and Tables

out-of-scope

manual page unknown

Appendix: Symbol Index

Symbol	Sections
AdvanceAfterCombat	§6.82, §7.6
AdvancePhase	§4
AngloEgyptianDirectFire	§6.24
AngloEgyptianStandard	§7.7
ArtilleryBreachWall	§6.63
BattalionOrdinal	§5.54
Breach	§6.63
BrigadeId	§2.3, §5.54
BrigadeIntegrity	§5.54, §6.24
BritishLeader	§6.51
CAMPAIGN_TURN_TRACK	§9.12
CampaignVictoryLevel	§9.14
CombatResult	§CRT
ConstructZariba	§5.3
DayNight	§8.1
Demolition	§6.53
DemolitionTarget	§6.53
DervishDesertion	§8.2
DervishGunboat	§2.32
DervishLeaderCommandMismatch	§5.53
DervishStandard	§7.7
DervishTribe	§7.4
DervishTribeMix	§5.52
DervishVsTrenchedDefender	§7.7
DieModifier	§7.7
DirectFire	§6.41
FALL_OF_KHARTOUM_SETUP	§9.321, §9.344
FALL_OF_KHARTOUM_TURN_TRACK	§9.321, §9.341
Faction	§9.112
FireAttack	§6.54
FireFactor	§6.11
FireFactorRow	§CRT
FoKVictoryLevel	§9.35
Fort	§5.25, §6.54
Friendlies	§6.52
FriendliesAction	§5.21
FriendliesTransport	§5.21
GameState	§4
GameTime	§9.12
GameTurnIndex	§4

GunboatId	§2.32, §6.64
GunboatMovement	§5.24
GunboatStack	§5.51
HISTORICAL_TURN_TRACK	§9.22
HexDirection	§5.11
HexDistance	§6.22, §7.5
HexsideKind	§5.23
HexsideRef	§5.23
HistoricalVictoryLevel	§9.24
HowitzerFire	§6.64
HowitzerResolution	§2.31, §6.64
Immobile	§5.25
Khor	§5.44
Location	§9.112
LosCondition	§6.3
LosFeature	§6.3
LosLevel	§6.3
MAX_CHAIN_HEXES	§10.2
MaximSecondAndHowitzer	§6.42
MeleeAttack	§7.7
MeleeCombat	§7.3
MeleeFactor	§7.1
MeleeModifier	§7.7
MineResult	§10.1
MovementAllowance	§5.11
MovementPoints	§5.11
NamedGunboat	§2.32
Old	§2.32
OldGunboat	§2.32
OptionalRule	§10
OverLimit	§5.51
PendingMelee	§4, §7
Phase	§4
PlaceReinforcements	§9.112
Range	§6.22
RangeBand	§6.16, §6.22
RetreatBeforeMelee	§7.5
RiverMine	§10.1, §10.12
RoyalEngineers	§6.53
ScatterDirection	§6.64
SetupLetter	§9.112
SpriteAnnotation	§2.3
Terrain	§6.23
TransportState	§5.21

TurnEntry	§9.12
TurnEvent	§8.2
TurnLabel	§9.12
UnitKind	§2.3, §7.4
UnitMovement	§5.11
UnitProfile	§2.3
UnitState	§6.54
VictoryLedger	§9.14
VictoryPoints	§9.14
VpEvent	§9.14
VpSource	§9.14
Wall	§5.44
WalledCityEntry	§5.23
WeaponClass	§2.31, §6.6, §6.61, §6.62
WhiteNileMouth	§9.345
Zariba	§5.44
ZaribaThornHedge	§9.23, §9.231
ZaribaTrench	§9.23, §9.232
ZaribaTrenchEntrenched	§9.23, §9.232
ZocReason	§5.41, §5.44, §6.54
advance_phase	§4
ae_range_effects	§6.22
ali_wad_helu	§9.322
anglo_egyptian_campaign_schedule	§9.113
apply_advance_after_combat	§6.82, §7.6
apply_artillery_breach_wall	§6.63
apply_construct_zariba	§5.3
apply_demolition	§6.53
apply_friendlylies_transport	§5.21
apply_howitzzer_fire	§6.64
apply_melee_combat	§7.3
apply_move_unit	§9.233
apply_place_chain	§10.2, §10.21
apply_place_mine	§10.11
apply_place_reinforcements	§9.112
apply_retreat_before_melee	§7.5
apply_river_mine	§10.1, §10.12, §10.13, §10.14
apply_sink_chain	§10.23
blocking_rules	§6.21, §6.3
blocks_advance_after_combat	§6.82
blocks_los	§6.3
blocks_melee	§7.2
blocks_movement	§5.23, §9.233
blocks_zoc	§5.44

brigade_integrity	§5.54
can_advance_after_combat	§6.7, §6.82, §7.6
can_fire_at	§6.64
can_fire_at_wall	§6.63
can_melee	§7.2, §7.4
can_move_gunboat	§10.22
can_move_unit_to	§5.22, §5.26, §5.43
can_place_chain	§10.21
can_retreat_before_melee	§7.5
check_gordon_palace	§9.346
check_invariants	§5.51, §5.52
check_stacking	§5.51
combat_results_table	§CRT
constructing_zariba	§5.3
defense_modifier	§6.23
demolishing	§6.53
dervish_campaign_schedule	§9.112
dervish_leader_for_setup_letter	§9.212
dervish_range_effects	§6.22
dervish_tribe	§2.31
die_modifier	§6.24
effective_movement_at_night	§8.1
end_player_turn	§4, §5.13
fall_of_khartoum_map_data	§9.342
fires_twice	§6.42
first_player	§9.211
from_superiority	§9.14
from_total	§CRT
halve	§8.1
has_combat_factors	§6.51
has_los	§6.21, §6.3
hex_has_enemy_fort	§9.344
hex_in_enemy_zoc	§5.26, §5.43, §5.44
hit_target_hex	§6.64
howitzer_scatter	§6.64
in_deployment_zone	§5.22, §9.212
is_boat	§5.24
is_crossroad	§5.11
is_friendlylies	§5.21, §6.52
is_gordon	§9.346
is_los_trees	§6.3
is_nile_mouth_crossing	§9.345
is_walled_city	§5.23
khalifa_abdullah	§2.31

loaded_on	§5.21
los_level	§6.3
los_level_for_unit	§6.3
may_act	§6.54
may_attack_this_turn	§5.3
may_be_melee_attacked	§7.1
may_enter_walled_city	§5.23
may_melee_attack	§7.4
may_retreat_before_melee	§7.5
mines	§10.11
movement_cost	§5.11
movement_cost_for	§5.42, §9.233
movement_cost_with_road	§5.11
mp_spent	§5.12
net_modifier	§6.54
new	§4
night_max_range	§8.1
passable_by_land	§5.11
points	§9.14
range_band_for	§9.343
roads	§5.11
score_elimination	§9.14
sections_for_picker	§9.322
setup_complete	§9.111
setup_letter	§9.212
sum	§7.1
sum_to_row	§6.14
superiority	§9.14
terrain_effects_chart	§5.11
total_for	§9.14
turn_marker_pixel	§9.12
unit_projects_zoc	§5.41, §5.44
value	§4
who_scores	§9.14
zariba_entry_surcharge	§9.233
zoc_hexes	§5.41